System I

Computational Operations & Units

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Overview

- Addition
- Subtraction
- Arithmetic logic unit (ALU)
- Multiplication
- Division
- Floating number operations

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Iterative Combinational Circuits

- Arithmetic functions
  - Operate on binary vectors
  - Use the same subfunction in each bit position
- Can design functional block for subfunction and repeat to obtain functional block for overall function
- *Cell* - subfunction block
- *Iterative array* - a array of interconnected cells
- An iterative array can be in a single dimension (1D) or multiple dimensions

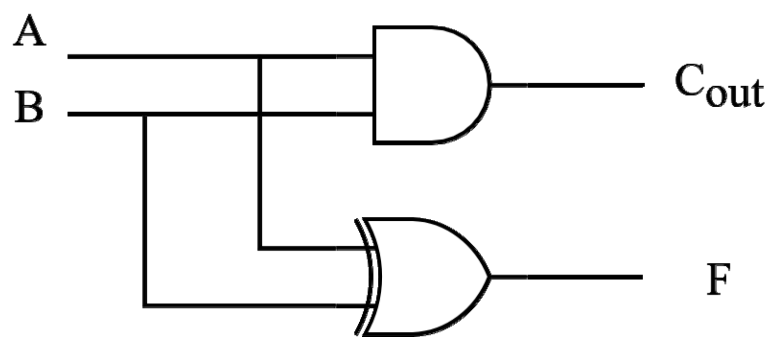
1-bit adder: Half-Adder

- A 2-input, 1-bit width binary adder that performs the following computations:

A	0	0	1	1
+ B	+ 0	+ 1	+ 0	+ 1
C F	0 0	0 1	0 1	1 0

- A half adder adds two bits to produce a two-bit sum
- The sum is expressed as a <u>sum bit</u> , F and a <u>carry bit</u>, C
- The half adder can be specified as a truth table for F and C ⇒

A	B	F	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1



1-bit adder: Full Adder

- A full adder is similar to a half adder, but includes a carry-in bit from lower stages.
- Like the half-adder, it computes a sum bit **F** and a carry bit $\mathbf{C_{in}}$
  - For a carry-in (C_{in}) of 0, it is the same as the half-adder:

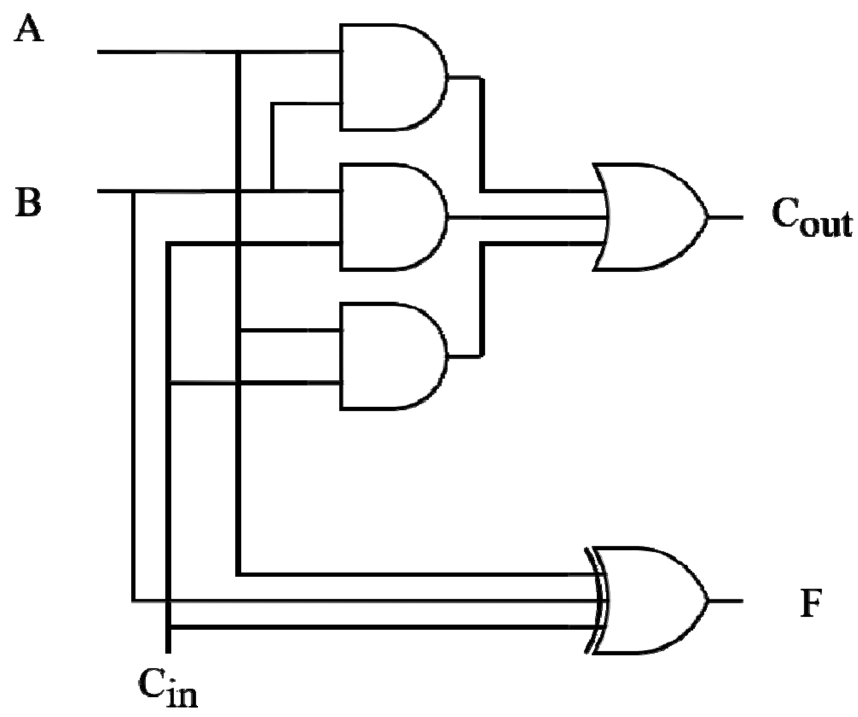
$\mathbf{C_{in}}$	**0**	**0**	**0**	**0**
A	**0**	**0**	**1**	**1**
+ B	**+ 0**	**+ 1**	**+ 0**	**+ 1**
$\mathbf{C_{out}}$ **F**	**0 0**	**0 1**	**0 1**	**1 0**

  - For a carry-in (C_{in}) of 1:

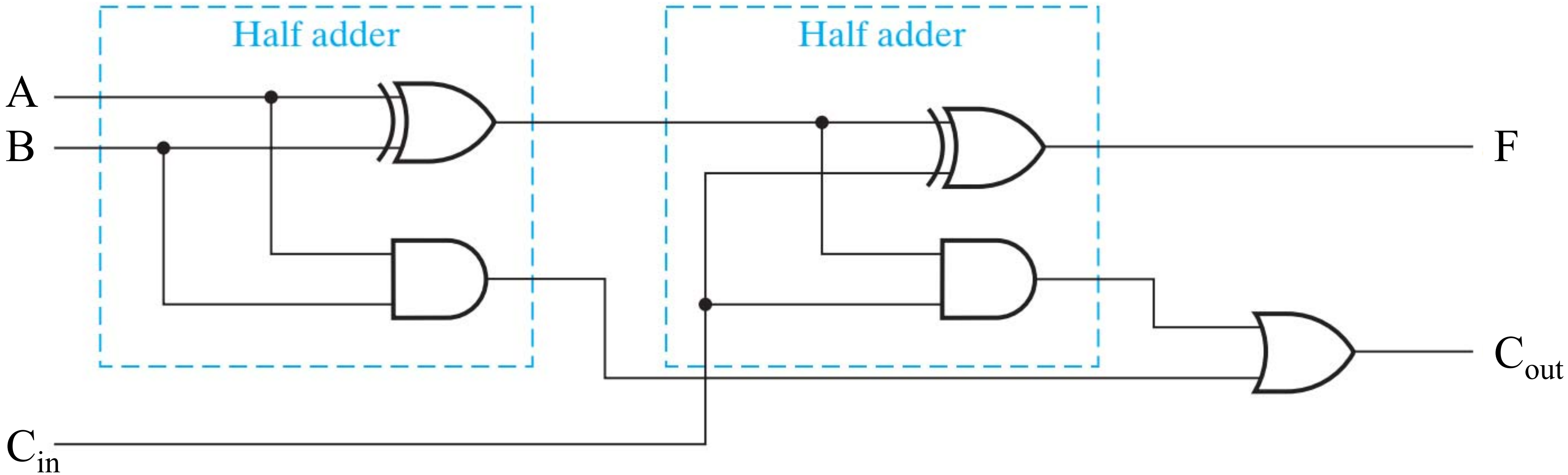
$\mathbf{C_{in}}$	**1**	**1**	**1**	**1**
A	**0**	**0**	**1**	**1**
+ B	**+ 0**	**+ 1**	**+ 0**	**+ 1**
$\mathbf{C_{out}}$ **F**	**0 1**	**1 0**	**1 0**	**1 1**

Classic Designs: Full Adder (cont'd)

Input			**Output**	
A	B	C_{in}	F	C_{out}
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

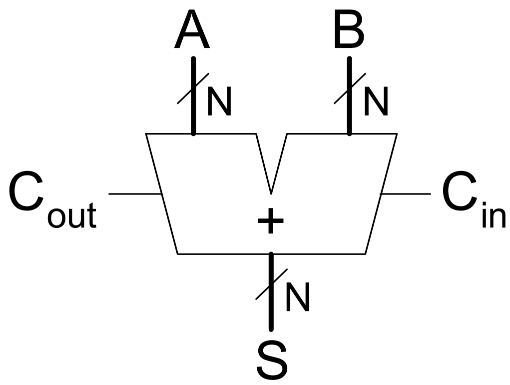


Classic Designs: Full Adder (cont'd)



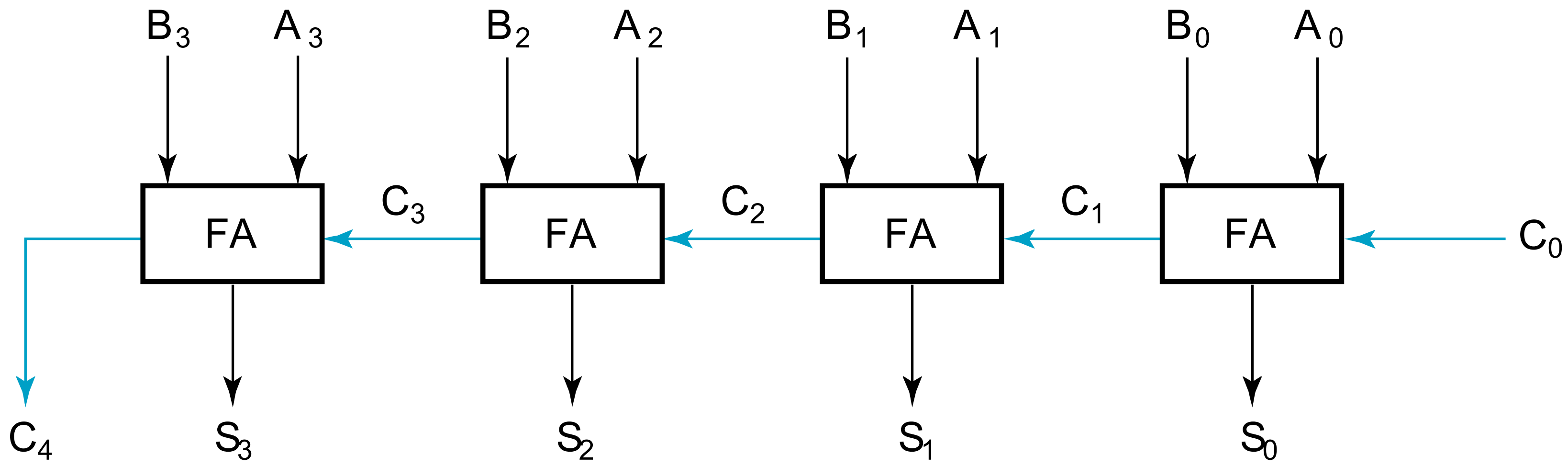
Multibit Carry Propagate Adders (CPA)

- Types of CPA
  - Ripple-carry (slow)
  - Carry Skip
  - Carry Select
  - Carry-lookahead (fast)
  - Prefix (fast)
- Faster adders require more hardware



Ripple-Carry Adder (RCA)

- Chain 1-bit adders together
- Carry ripples through entire chain
- Disadvantage: **slow**



Ripple-Carry Adder Delay

$$t_{ripple} = Nt_{FA}$$

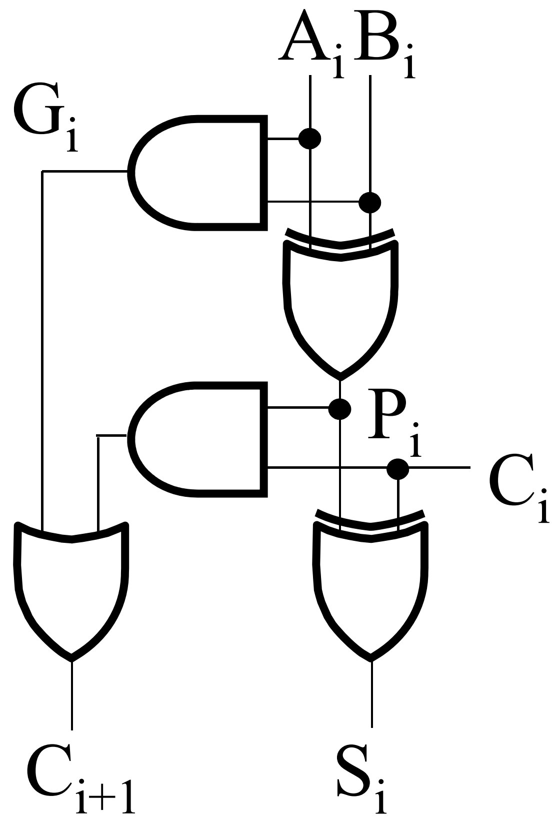
t_{FA}: delay of a 1-bit full adder

Carry Lookahead Adder

$$P_i = A_i \oplus B_i \qquad G_i = A_i B_i$$

$$S_i = P_i \oplus C_i \qquad C_{i+1} = G_i + P_i C_i$$

- For a given i-bit binary adder,
  - If $A_i = B_i =$ "1" and whatever C_i is, we have carry out as 1, that is $C_{i+1} = 1$
  - If the output of half adder is 1 and we have carry in as 1, we have carry out as 1， that is $C_{i+1} = 1$
- These two conditions of setting carry out as 1 is called *generate* (G_i) and *propagate* (P_i).



Carry Lookahead Development

- $C_{i+1} = G_i + P_iC_i$ can be removed from the cells and used to derive a set of carry equations spanning multiple cells.
- Beginning at the cell 0 with carry in C_0:

$$\mathbf{C_1 = G_0 + P_0\, C_0}$$

$$\mathbf{C_2 = G_1 + P_1\, C_1 = G_1 + P_1(G_0 + P_0\, C_0)}$$

$$\mathbf{= G_1 + P_1G_0 + P_1P_0\, C_0}$$

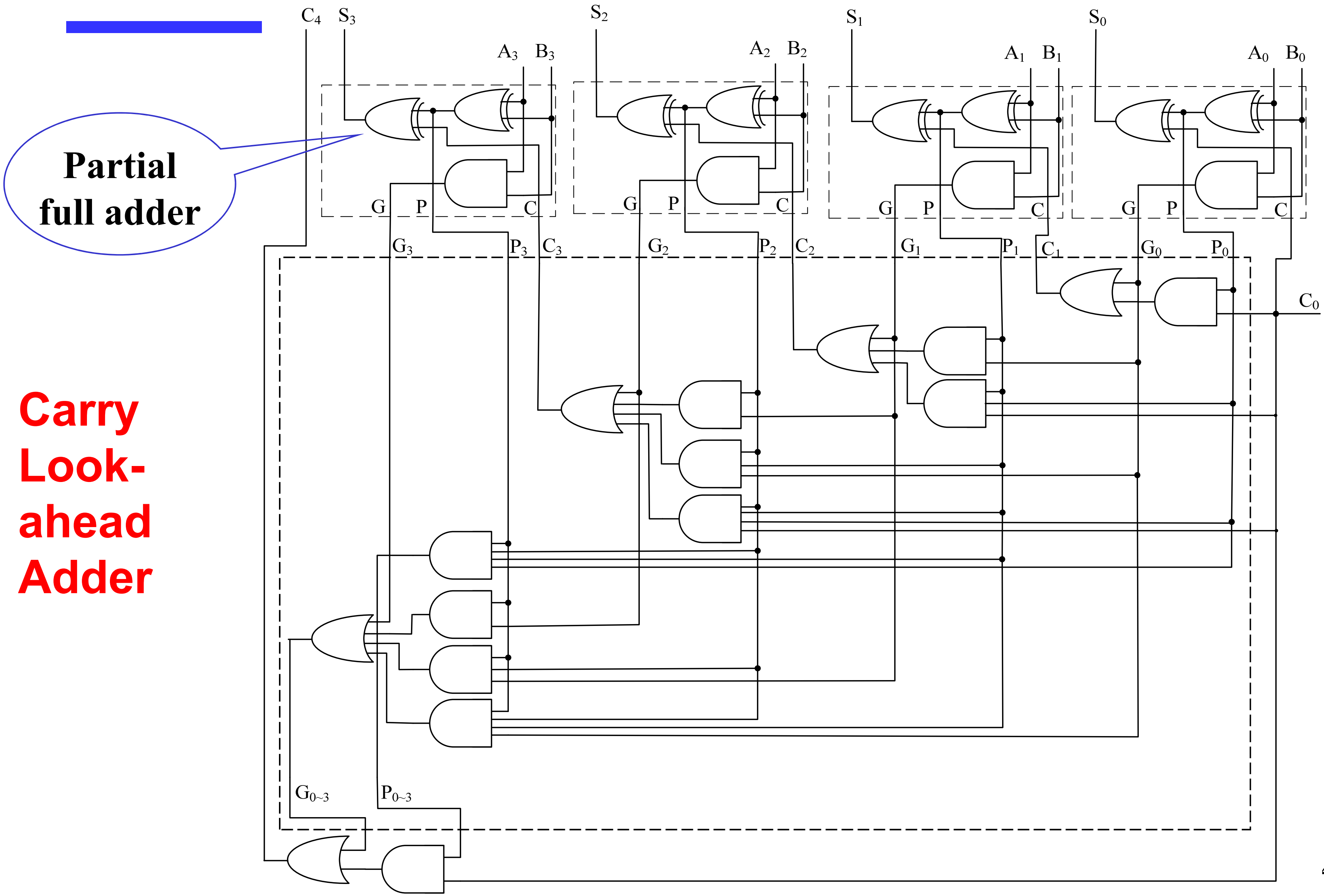
$$\mathbf{C_3 = G_2 + P_2\, C_2 = G_2 + P_2(G_1 + P_1G_0 + P_1P_0\, C_0)}$$

$$\mathbf{= G_2 + P_2G_1 + P_2P_1G_0 + P_2P_1P_0\, C_0}$$

$$\mathbf{C_4 = G_3 + P_3\, C_3}$$

$$\mathbf{= G_3 + P_3G_2 + P_3P_2G_1 + P_3P_2P_1G_0 + P_3P_2P_1P_0\, C_0}$$

4-Bit CLA



Carry Look-ahead Adder

Group Carry Lookahead Logic

- Figure in the previous slide shows the implementation of these equations for four bits. This could be extended to more than four bits; in practice, due to limited gate fan-in, such extension is not feasible.
- $C_4 = G_3 + P_3G_2 + P_3P_2G_1 + P_3P_2P_1G_0 + P_3P_2P_1P_0\, C_0$
- Instead, the concept is extended another level by considering *group generate* ($G_{0\text{-}3}$) and *group propagate* ($P_{0\text{-}3}$) functions:

$$G_{0\sim3} = G_3 + P_3\, G_2 + P_3\, P_2\, G_1 + P_3\, P_2\, P_1\, P_0\, G_0$$
$$P_{0\sim3} = P_3\, P_2\, P_1\, P_0$$

- Using these two equations:

$$C_4 = G_{0\sim3} + P_{0\sim3}\, C_0$$

- Thus, it is possible to have four 4-bit adders use one of the same carry lookahead circuit to speed up 16-bit addition

Group Carry Lookahead Logic (Cont.)

- $C_4 = G_3 + P_3G_2 + P_3P_2G_1 + P_3P_2P_1G_0 + P_3P_2P_1P_0C_0 = G_{0\sim3} + P_{0\sim3}C_0$
- $C_8 = G_7 + P_7G_6 + P_7P_6G_5 + P_7P_6P_5G_4 + P_7P_6P_5P_4C_4 = G_{4\sim7} + P_{4\sim7}C_4$
- $C_{12} = G_{11} + P_{11}G_{10} + P_{11}P_{10}G_9 + P_{11}P_{10}P_9G_8 + P_{11}P_{10}P_9P_8C_8 = G_{8\sim11} + P_{8\sim11}C_8$
- $C_{16} = G_{15} + P_{15}G_{14} + P_{15}P_{14}G_{13} + P_{15}P_{14}P_{13}G_{12} + P_{15}P_{14}P_{13}P_{12}C_{12} = G_{12\sim15} + P_{12\sim15}C_{12}$

Group Carry Lookahead Logic (Cont.)

- $C_4 = G_{0\sim3} + P_{0\sim3}C_0$
- $C_8 = G_{4\sim7} + P_{4\sim7}C_4$
- $C_{12} = G_{8\sim11} + P_{8\sim11}C_8$
- $C_{16} = G_{12\sim15} + P_{12\sim15}C_{12}$

$C_1 = G_0 + P_0 C_0$

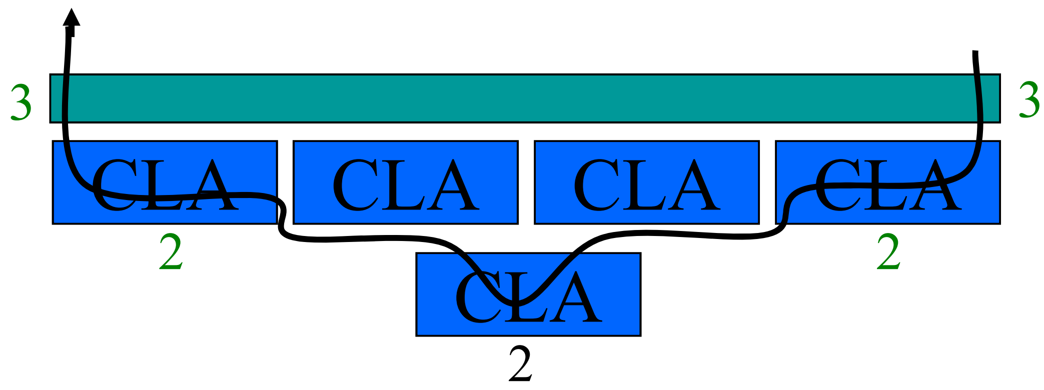
$C_2 = G_1 + P_1 C_1$

$C_3 = G_2 + P_2 C_2$

$C_4 = G_3 + P_3 C_3$

Carry Lookahead Example

- Specifications:
  - 16-bit CLA
  - Delays:
    - NOT = 1
    - XOR = Isolated AND = 3
    - AND-OR = 2



- Longest Delays:
  - Ripple carry adder* = 3 + 15 × 2 + 3 = 36
  - CLA = 3 + 3 × 2 + 3 = 12

Prefix Adder

- Computes carry in (C_{i-1}) for each column, then computes sum:

$$\mathbf{S_i = (A_i \oplus B_i) \oplus C_{i-1}}$$

- Computes G and P for 1-, 2-, 4-, 8-bit blocks, etc. until all G_i (carry in) known

- $\log_2 N$ stages

Prefix Adder (cont'd)

- Carry in either *generated* in a column or *propagated* from a previous column.
- Column -1 holds C_{in}, so

  $\mathbf{G_{-1} = C_{in}, P_{-1} = 0}$

- Carry in to column i == carry out of column i-1:

  $\mathbf{C_{i-1} = G_{i-1:-1}}$

  $G_{i-1:-1}$: generate signal spanning columns i-1 to -1

- Sum equation:

  $\mathbf{S_i = (A_i \oplus B_i) \oplus G_{i-1:-1}}$

- **Goal**: Quickly compute $G_{0:-1}$, $G_{1:-1}$, $G_{2:-1}$, $G_{3:-1}$, $G_{4:-1}$, $G_{5:-1}$, … (called ***prefixes***)

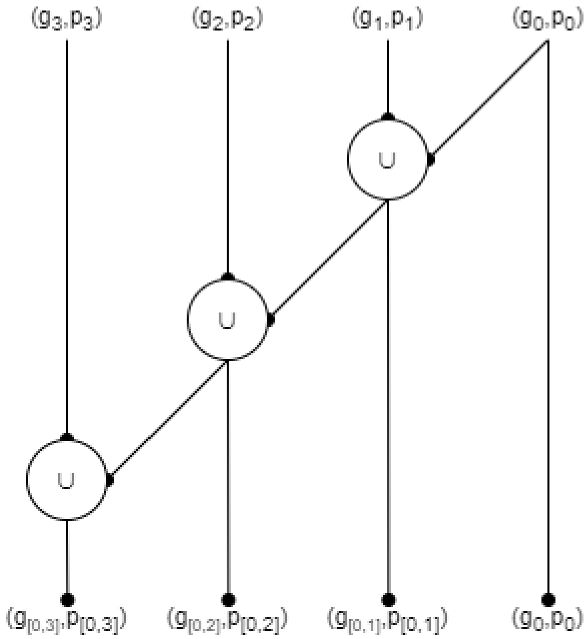
Prefix Adder (cont'd)

- Generate and propagate signals for a block spanning bits i:j:

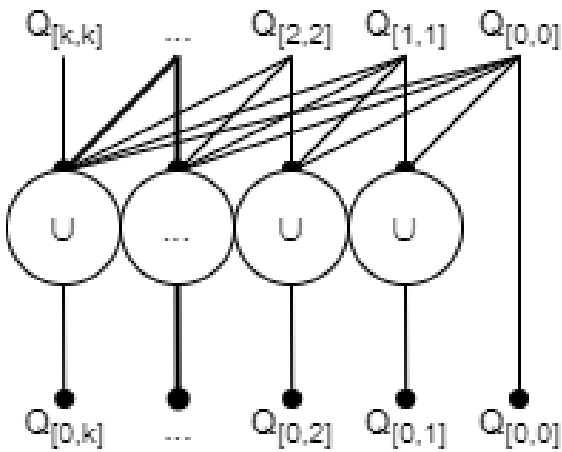
  $$\mathbf{G_{i:j} = G_{i:k} + P_{i:k}\, G_{k-1:j}}$$

  $$\mathbf{P_{i:j} = P_{i:k} P_{k-1:j}}$$

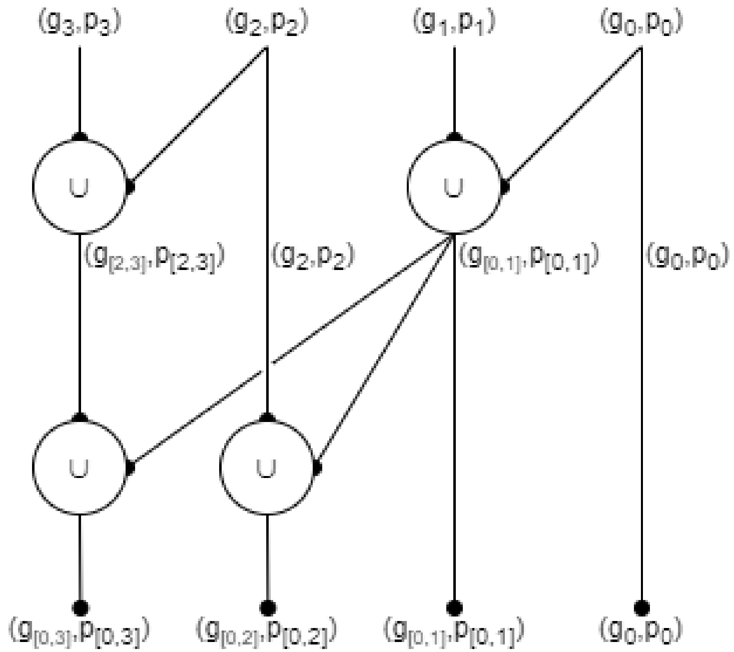
- In words:
  - **Generate**: block i:j will generate a carry if:
    - upper part (i:k) generates a carry or
    - upper part propagates a carry generated in lower part (k-1:j)
  - **Propagate**: block i:j will propagate a carry if *both* the upper and lower parts propagate the carry



(g3,p3)
(g2,p2)
(g1,p1)
(g0,p0)
∪
∪
∪
(g[0,3],p[0,3])
(g[0,2],p[0,2])
(g[0,1],p[0,1])
(g0,p0)

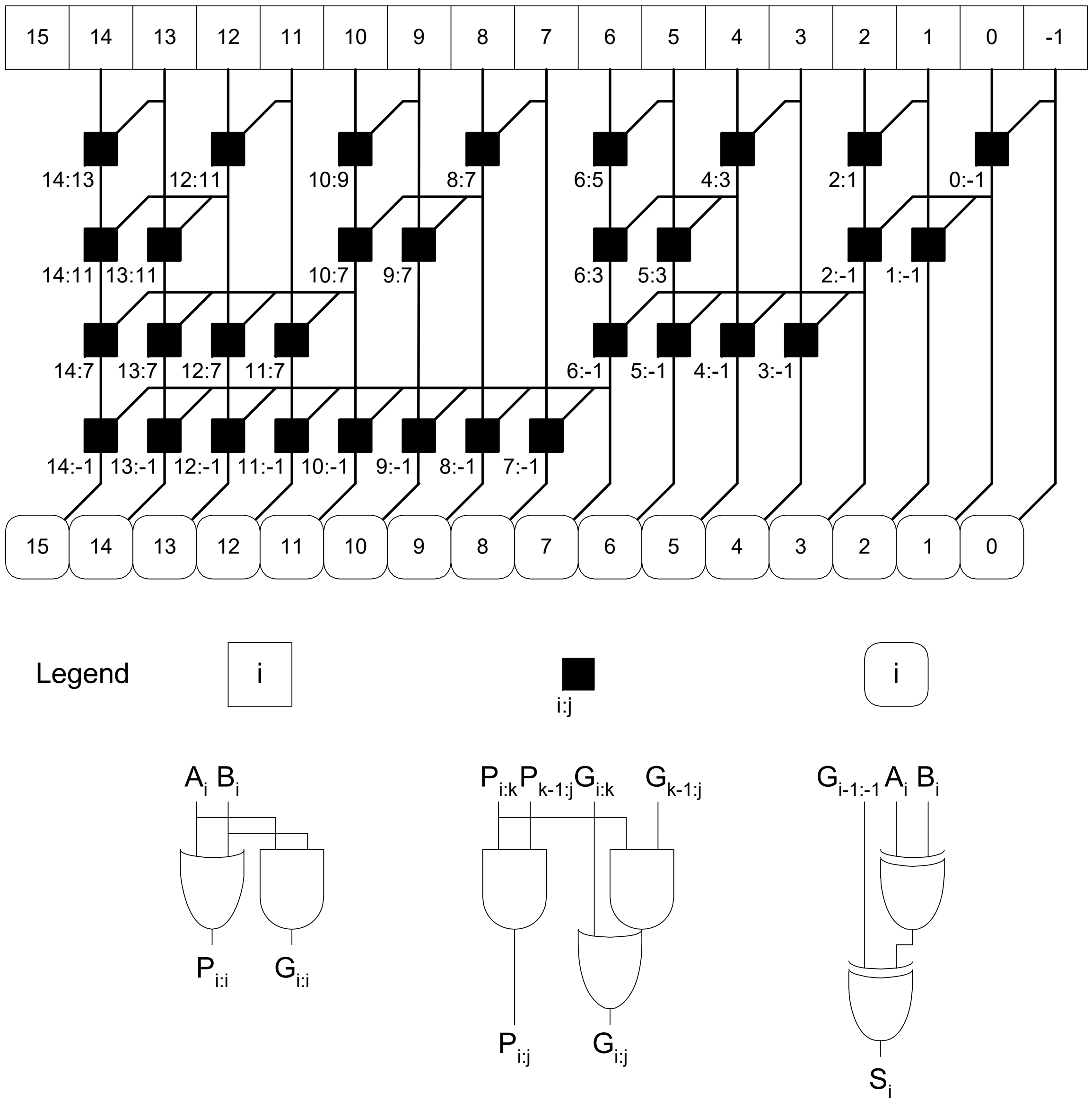


Q[k,k]
...
Q[2,2]
Q[1,1]
Q[0,0]
∪
...
∪
∪
Q[0,k]
...
Q[0,2]
Q[0,1]
Q[0,0]



(g3,p3)
(g2,p2)
(g1,p1)
(g0,p0)
∪
∪
(g[2,3],p[2,3])
(g2,p2)
(g[0,1],p[0,1])
(g0,p0)
∪
∪
(g[0,3],p[0,3])
(g[0,2],p[0,2])
(g[0,1],p[0,1])
(g0,p0)

Prefix Adder Schematic



Prefix Adder Delay

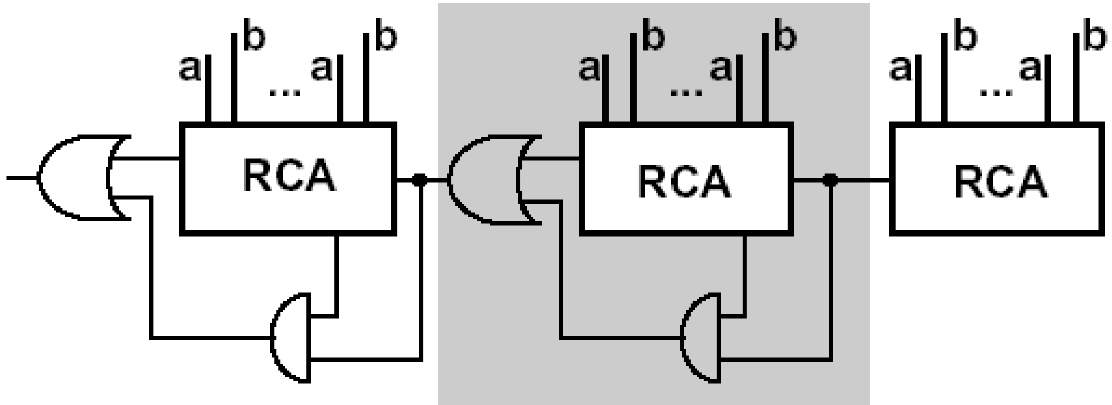
$$t_{PA} = t_{pg} + \log_2 N(t_{pg\_prefix}) + t_{XOR}$$

t_{pg}: delay to produce P_iG_i (AND or OR gate)

t_{pg_prefix}: delay of black prefix cell (AND-OR gate)

Carry skip adder

- Accelerating the carry by skipping the interior blocks
- Optimal speed with no-equal distribution of block length



Carry skip adder

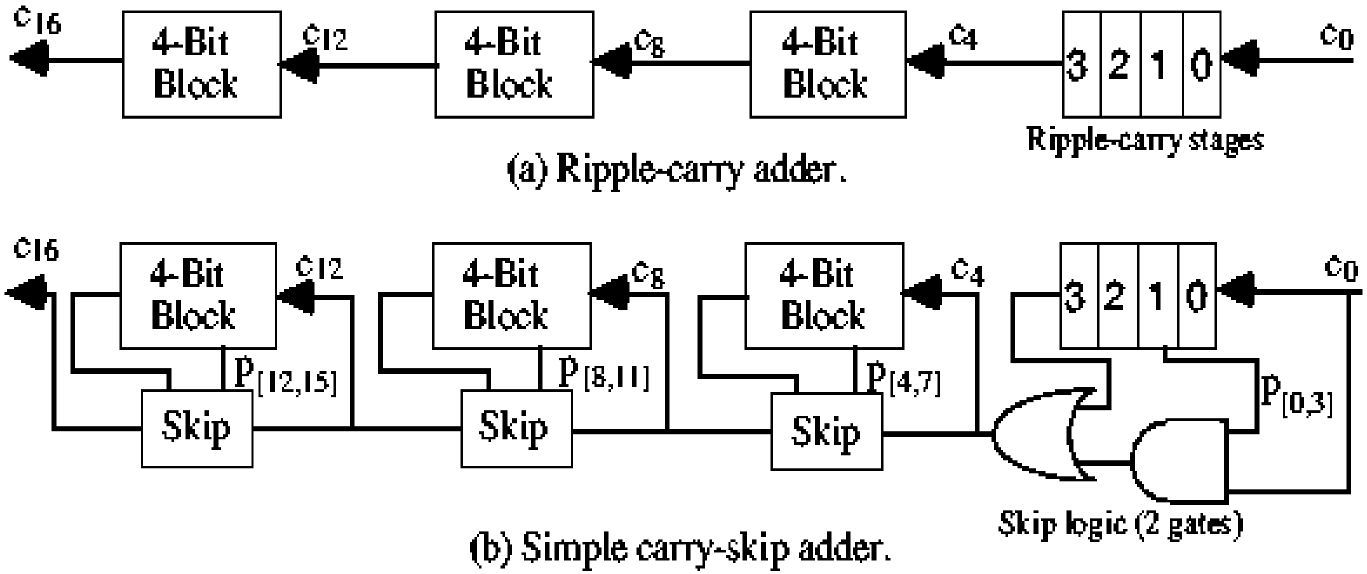
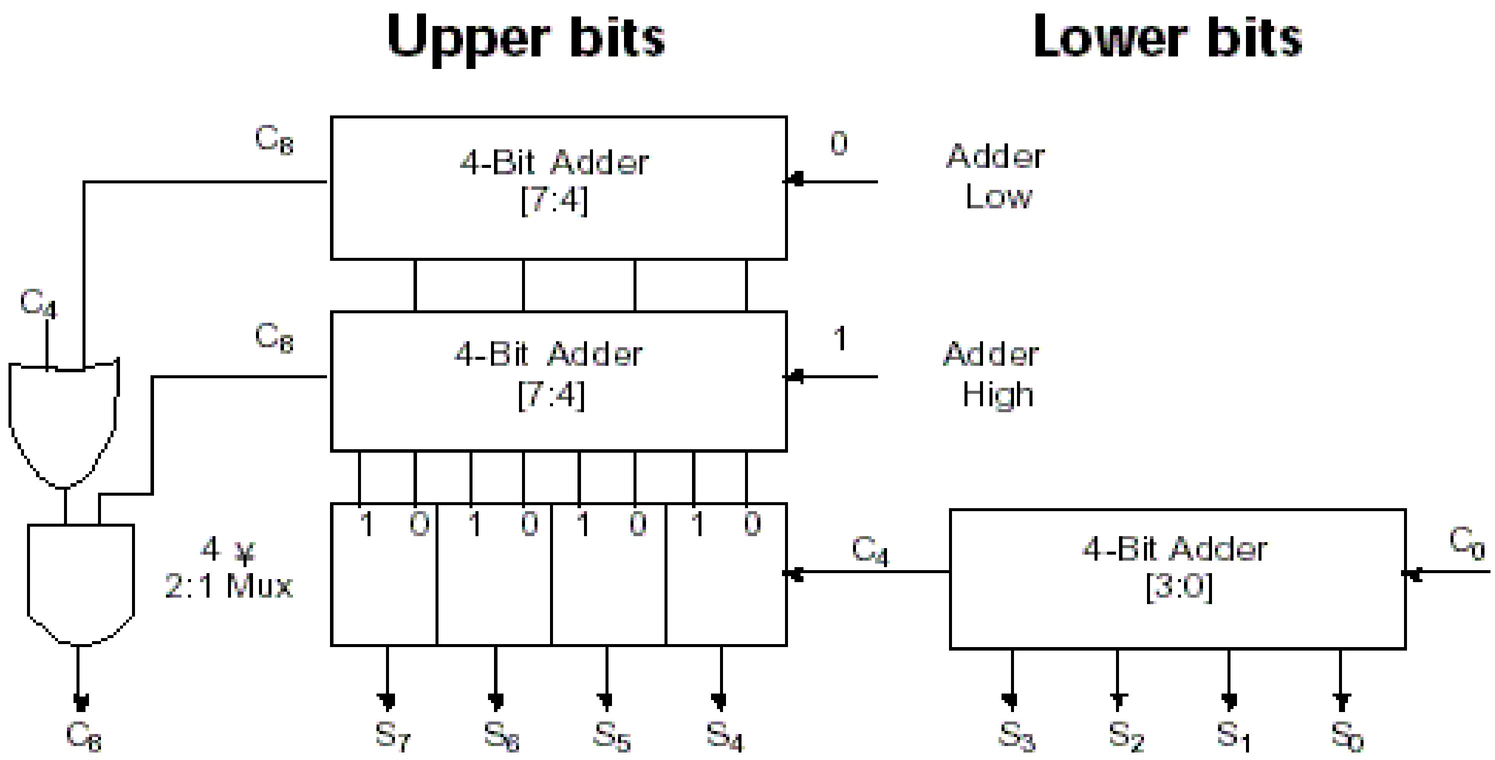


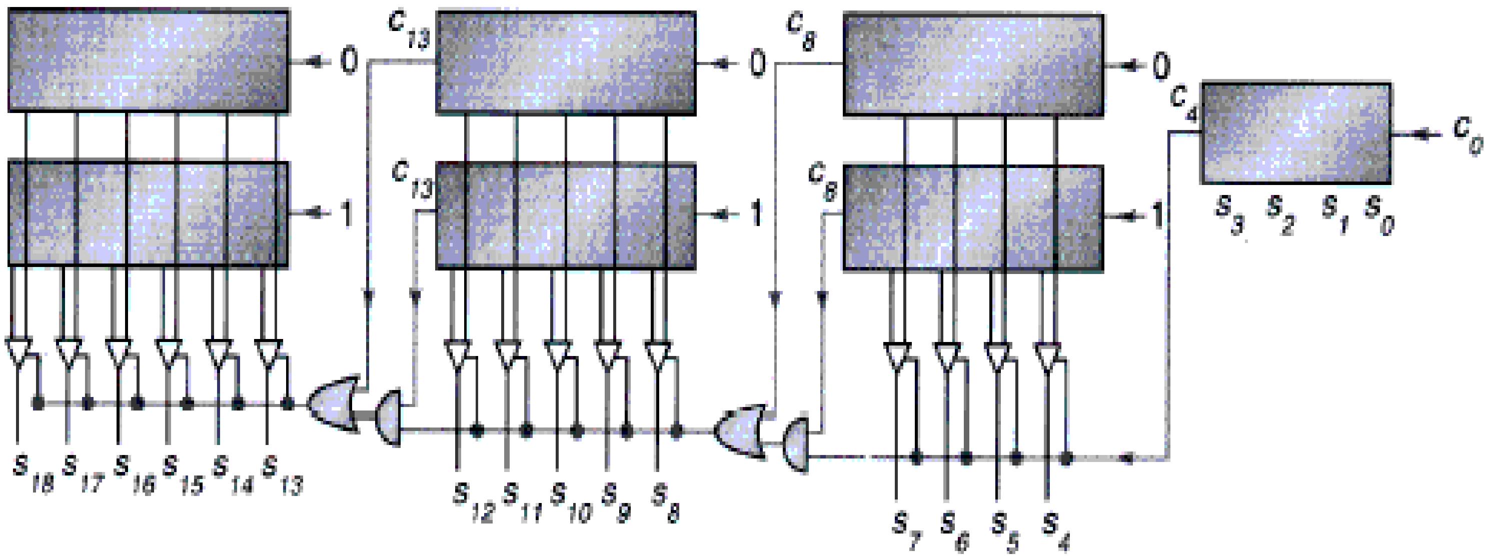
Fig. 7.1 **Converting a 16-bit ripple-carry adder into a simple carry-skip adder with 4-bit skip blocks.**

Carry select adder (CSA)



Carry select adder

- Carry selection by nibbles



Overview

- Addition
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- Arithmetic logic unit (ALU)
- Multiplication
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Unsigned Subtraction

- For n-digit, unsigned numbers M and N, find M – N in base 2:
  - Add the 2's complement of the subtrahend N to the minuend M:

    $$M + (2^n - N) = M - N + 2^n$$
  - If $M \geq N$, the sum produces end carry r^n which is discarded; from above, M – N remains.
  - If M < N, the sum does not produce an end carry and, from above, is equal to $2^n - (N - M)$, the 2's complement of (N – M).
  - To obtain the result – (N – M) , take the 2's complement of the sum and place a – to its left.

Unsigned 2's Complement Subtraction Example 1

- Find $01010100_2 - 01000011_2$

$$\begin{array}{r} 01010100 \\ -\ \underline{01000011} \end{array} \xrightarrow{\textbf{2's comp}} \begin{array}{r} \mathbf{1}\ 01010100 \\ +\ \underline{10111101} \\ 00010001 \end{array}$$

- The carry of 1 indicates that no correction of the result is required.

Unsigned 2's Complement Subtraction Example 2

- Find $01000011_2 - 01010100_2$

$$
\begin{array}{r}
01000011 \\
-\ \underline{01010100}
\end{array}
\xrightarrow{\text{2's comp}}
\begin{array}{r}
{}^{\mathbf{0}}\,01000011 \\
+\ \underline{10101100} \\
11101111
\end{array}
\xrightarrow{\text{2's comp}}
\begin{array}{r}
\\
\\
\\
00010001
\end{array}
$$

- The carry of 0 indicates that a correction of the result is required.
- Result = – (00010001)

Signed 2's Complement Arithmetic Examples

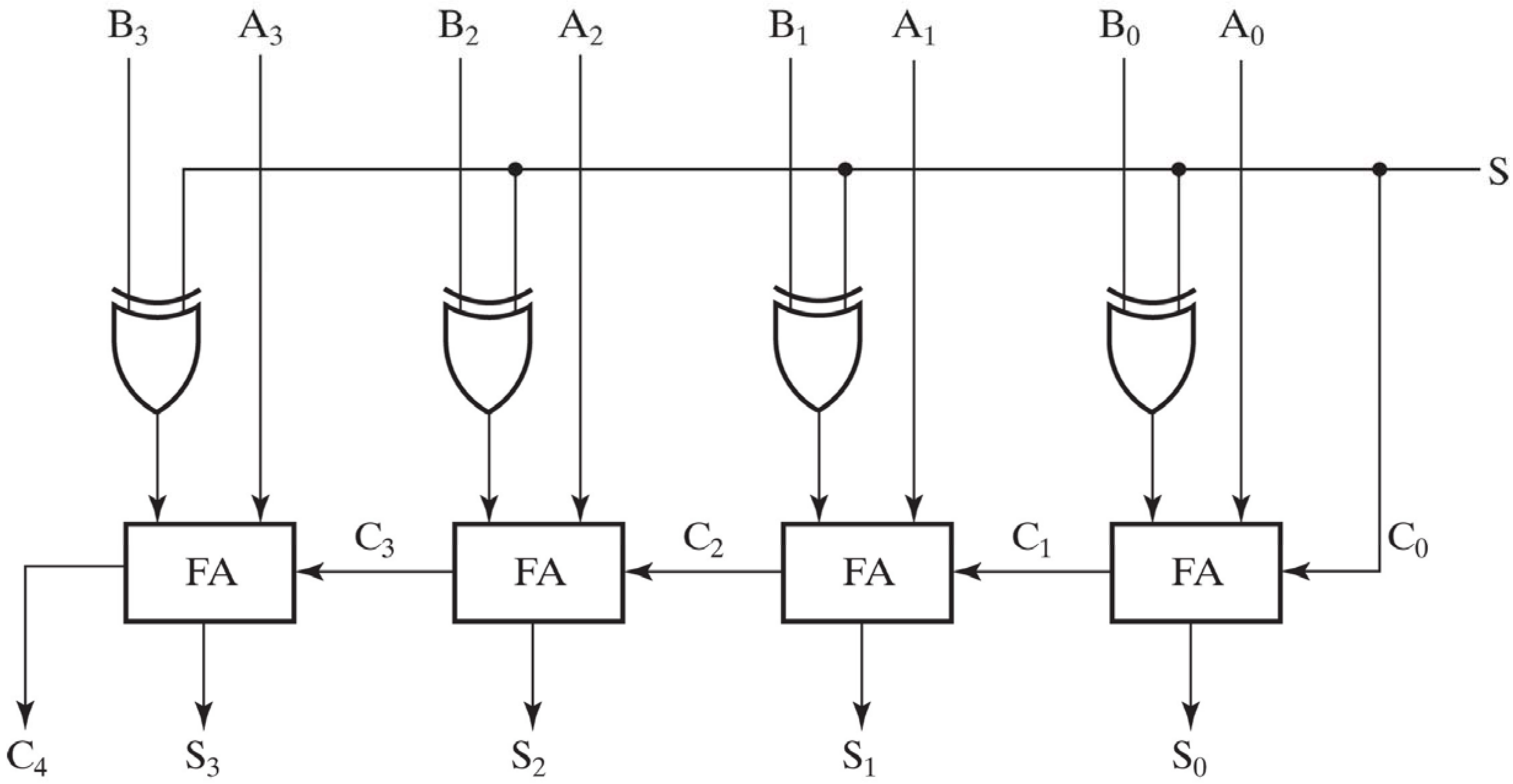
- Signed binary addition using 2s complement

+ 6	00000110	− 6	11111010	+ 6	00000110	− 6	11111010
+ 13	00001101	+ 13	00001101	− 13	11110011	− 13	11110011
+ 19	00010011	+ 7	00000111	− 7	11111001	− 19	11101101

- Signed binary subtraction using 2s complement

− 6	11111010	11111010	+ 6	00000110	00000110
− (−13)	− 11110011	+ 00001101	− (−13)	− 11110011	+ 00001101
+ 7		00000111	+ 19		00010011

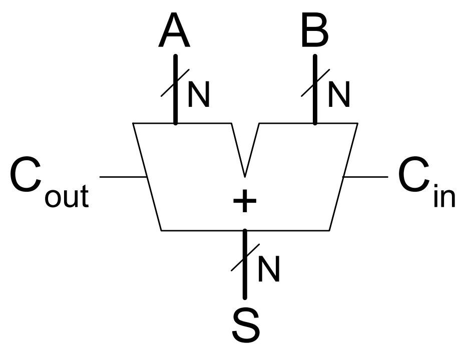
4-Bit Binary Adder-Subtractors

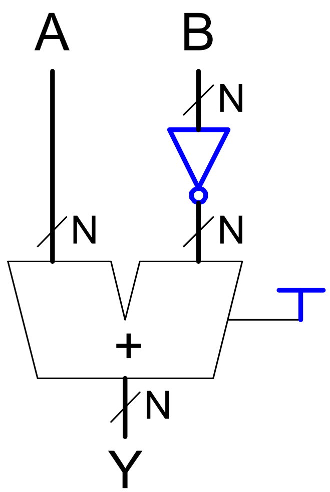


- When S=0: Addition (A+B)
- When S=1: Subtraction (A+2's complement of B)
- Can be used to add/subtract unsigned numbers and signed 2's complement numbers

Addition/Subtraction

- Both can be handled by using 2's complement representation
- Can achieve a unified implementation
  - Addition vs. subtraction
  - Unsigned vs. signed





- Corner cases shall be considered
  - Some important flags

Carry & Overflow

- Carry is important when…
  - Adding or subtracting ***unsigned integers***
  - Indicates that the unsigned sum is out of range
  - Either < 0 or > maximum unsigned n-bit value
- Overflow is important when…
  - Adding or subtracting ***signed integers***
  - Indicates that the signed sum is out of range

- Overflow occurs when?

Signed Overflow

- With two's complement and a 4-bit adder, for example, the largest representable decimal number is +7, and the smallest is -8.
- What if you try to compute 4 + 5, or (-4) + (-5)?

```
    0100   (+4)          1100   (-4)
  + 0101   (+5)        + 1011   (-5)
  -------------        -------------
   01001   (-7)         10111   (+7)
```

- We cannot just include the carry out to produce a five-digit result, as for unsigned addition. If we did, (-4) + (-5) would result in +23!
- Also, unlike the case with unsigned numbers, the carry out cannot be used to detect overflow, by itself
  - In the example on the left, the carry out is 0 but there is overflow.
  - Conversely, there are situations where the carry out is 1 but there is no overflow.

How to Detect Signed Overflow?

- The impact of carry and overflow

Expression	Result	Carry?	Overflow?	Correct Result?
0100(+4)+0010(+2)	0110(+6)	No	No	Yes
0100(+4)+0110(+6)	1010(-6)	No	Yes	No
1100(-4)+1110(-2)	1010(-6)	Yes	No	Yes
1100(-4)+1010(-6)	0110(+6)	Yes	Yes	No

Examples of four *signed* additions

- The easiest way to detect signed overflow is to look at all the sign bits.

```
   0100   (+4)          1100   (-4)
+  0101   (+5)       +  1011   (-5)
----------------     ----------------
  01001   (-7)         10111   (+7)
```

Detecting Signed Overflow

- Overflow occurs only in the two situations:
  - Adding two positive numbers and the sum is negative
  - Adding two negative numbers and the sum is positive
  - Can happen because of the fixed number of sum bits
- Overflow cannot occur if you add a positive number to a negative number. Do you see why?

- In two’s complement addition/subtraction
  - If the two numbers have the same sign bit and the sum/difference has a different sign bit, then overflow
  - Or, if the carry input and the carry output of the sign bit are different

Overflow Detection

- For unsigned number
  - Add
    - The carry of 1 indicates overflow
  - Subtraction
    - The carry of 1 indicates that no correction of the result is required
    - The carry of 0 indicates that a correction of the result is required

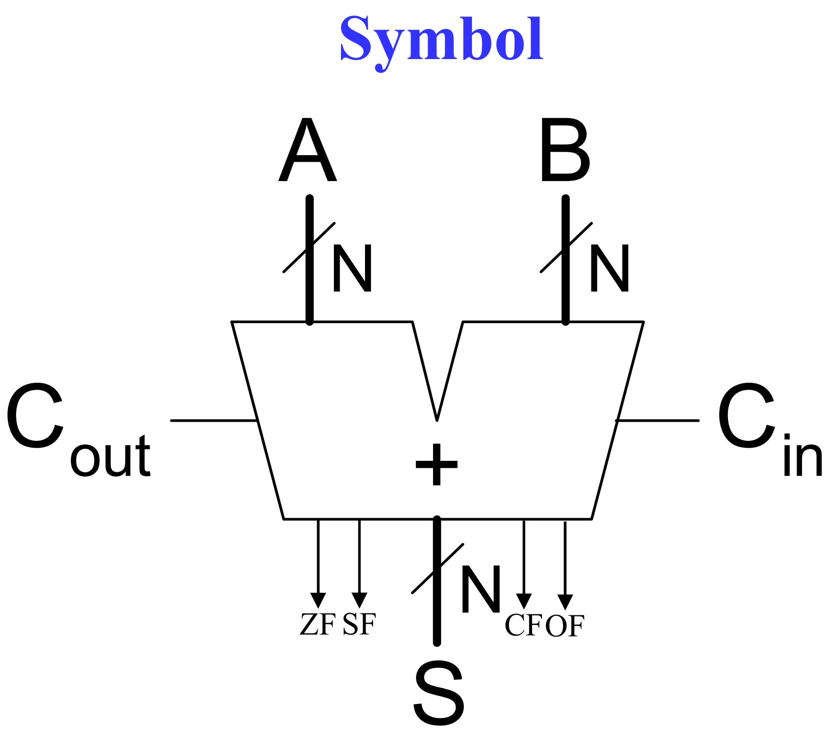
- For signed number
  - $V = C_n \oplus C_{n-1}$

Important Flags

- Zero flag (ZF)
  - ZF = 1 means the result is 0
  - Valid for both unsigned and signed operations
- Sign flag (SF/NF)
  - The sign of the result, i.e., S_{n-1}
  - Valid for signed operations
- Carry/borrow flag (CF)
  - If CF = 1
    - Carry for addition, i.e., C_{out}
    - Borrow for subtraction, i.e., $\sim C_{out}$
  - Valid for unsigned operations
- Overflow flag (OF)
  - Valid for signed operations

Adders with Flags

- ZF, SF, CF and OF



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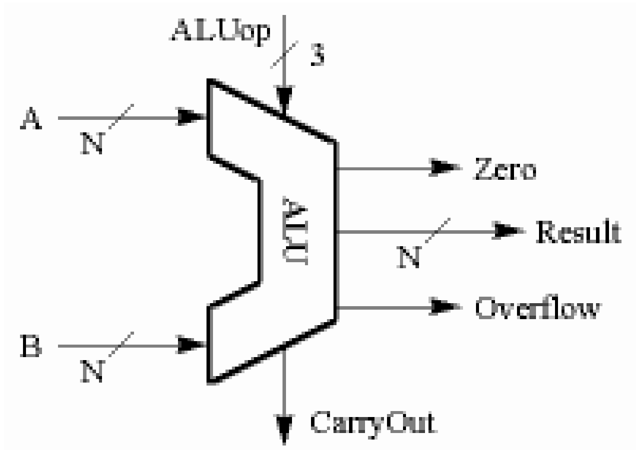
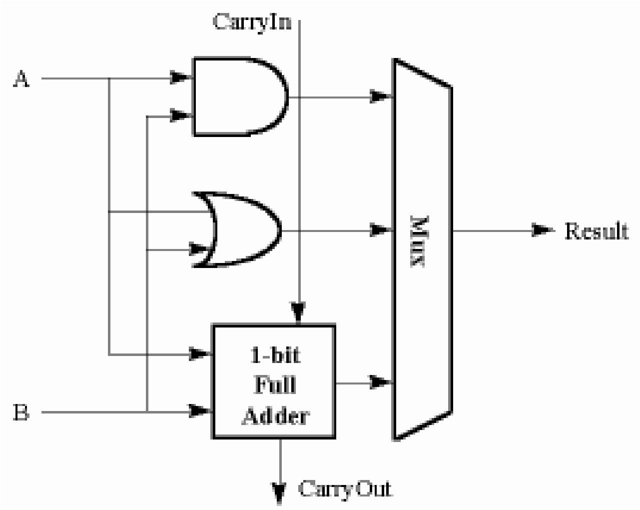
What is ALU?

- An arithmetic-logic unit, or ALU, performs many different arithmetic and logic operations. The ALU is the “heart” of a processor—you could say that everything else in the CPU is there to support the ALU.
- Two methods constitute the ALU
  - extended the adder
  - Parallel redundant select



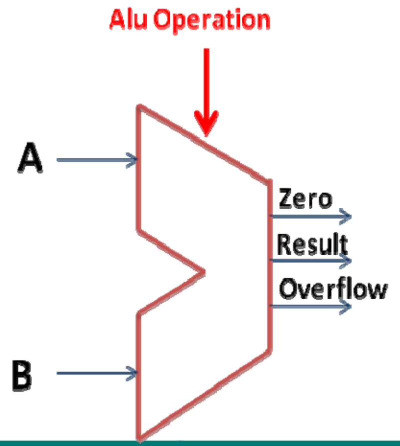
A Simple Example: 1-Bit ALU with 3 OPs

$F_{1:0}$	Function
00	A & B
01	A | B
10	A + B
11	not used

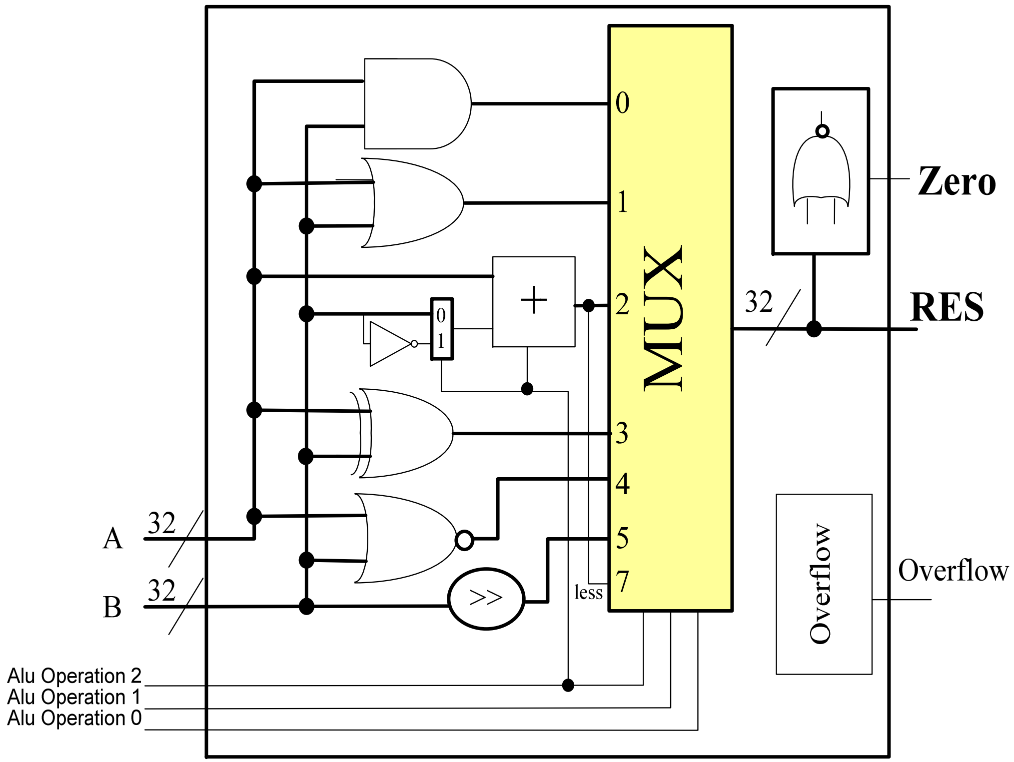



```
module ALU(
    input [3:0] a, b,
    input [1:0] aluop,
    input cin,
    output [3:0] f,
    output OF, SF, ZF, CF,
    output cout
);
    wire [3:0] sum;
    CLA_FLAGS(a, b, cin, sum, OF, SF, ZF,
CF, cout);
    always @(*) begin
        case(aluop)
            2'b00: f = a & b;
            2'b01: f = a | b;
            2'b10: f = sum;
            default: f = 0;
        endcase
    end
endmodule
```

N-Bit ALU with 8 OPs

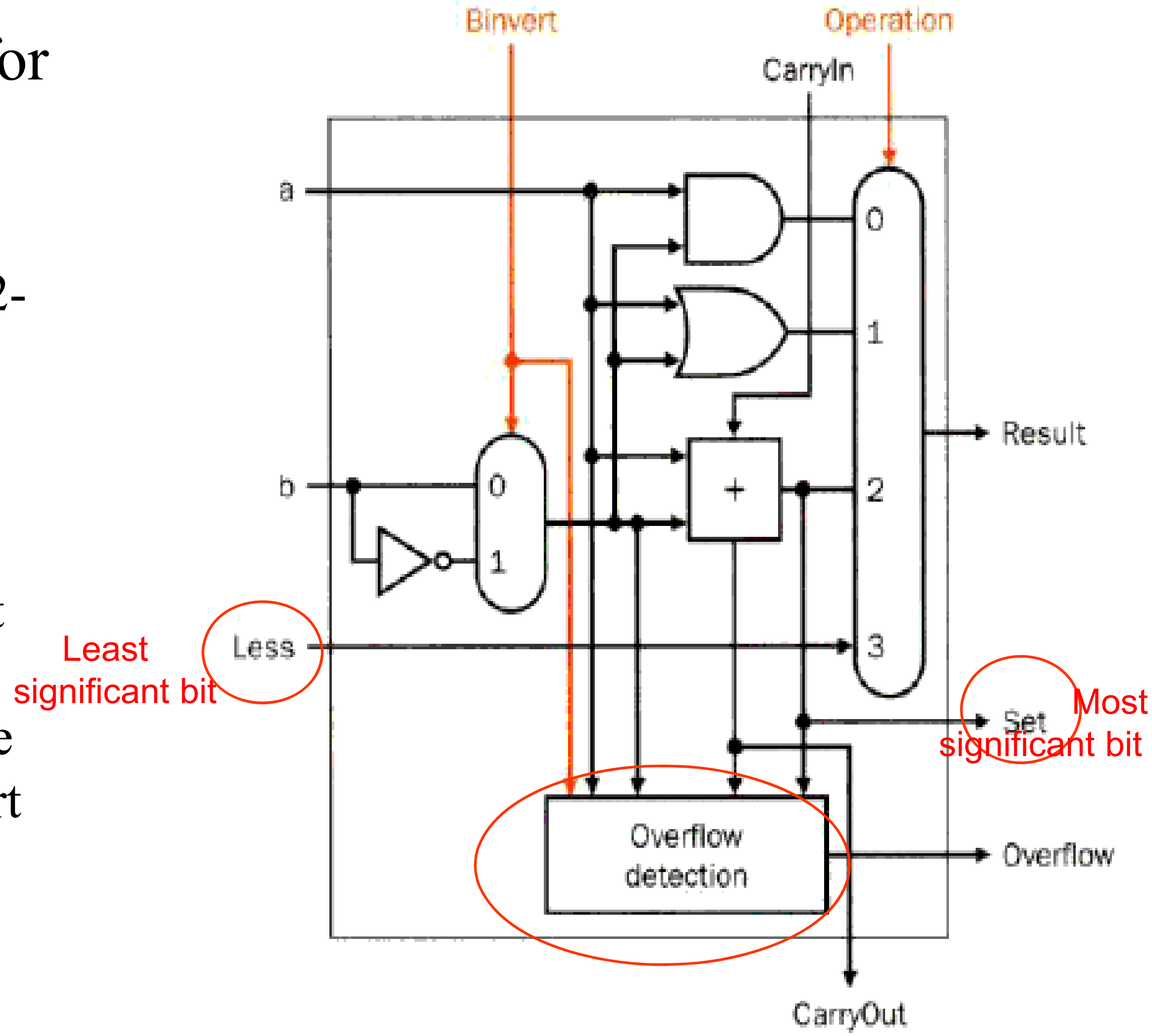


ALU Control Lines	Function
000	And
001	Or
010	Add
110	Sub
111	SLT
100	nor
101	srl
011	xor



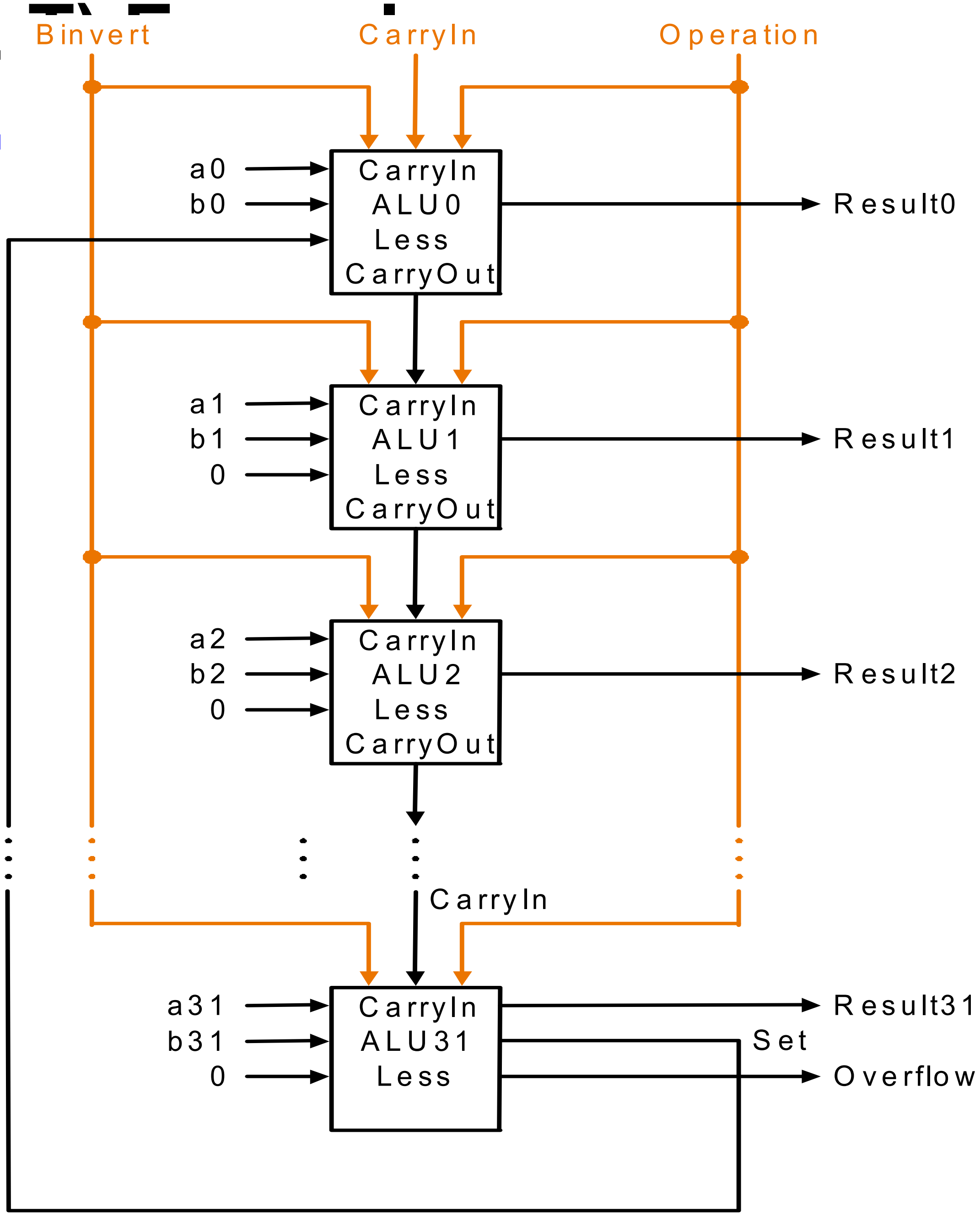
Set Less Than (SLT) Example

- Configure 32-bit ALU for SLT operation: A = 25 and B = 32
  - A < B, so Y should be 32-bit representation of 1 (0x00000001)
  - slt rd,rs,rt
  - If rs < rt, rd=1, else rd=0
  - For rd, all bits = 0 except the least significant
  - Subtraction (rs - rt), if the result is negative→ rs < rt
  - **Use of sign bit as indicator**



Set Less Than (SL

- Configure 32-bit ALU for SLT operation: A = 25 and B = 32
  - A < B, so Y should be 32-bit representation of 1 (0x00000001)
  - $F_{2:0}$ = 111
    - F_2 = 1 (adder acts as subtracter), so 25 - 32 = -7
    - -7 has 1 in the most significant bit (S_{31} = 1)
    - $F_{1:0}$ = 11 multiplexer selects Y = S_{31} (zero extended) = 0x00000001.



Various Use of Shifters

- Logical shifter: shifts value to left or right and fills empty spaces with 0's
  - 11001 >> 2 = 00110
  - 11001 << 2 = 00100
- Arithmetic shifter: same as logical shifter, but on right shift, fills empty spaces with the old most significant bit (MSB).
  - 11001 >>> 2 = 11110
  - 11001 <<< 2 = 00100
- Rotator: rotates bits in a circle, such that bits shifted off one end are shifted into the other end
  - 11001 ROR 2 = 01110
  - 11001 ROL 2 = 00111

Shifters as Multipliers and Dividers

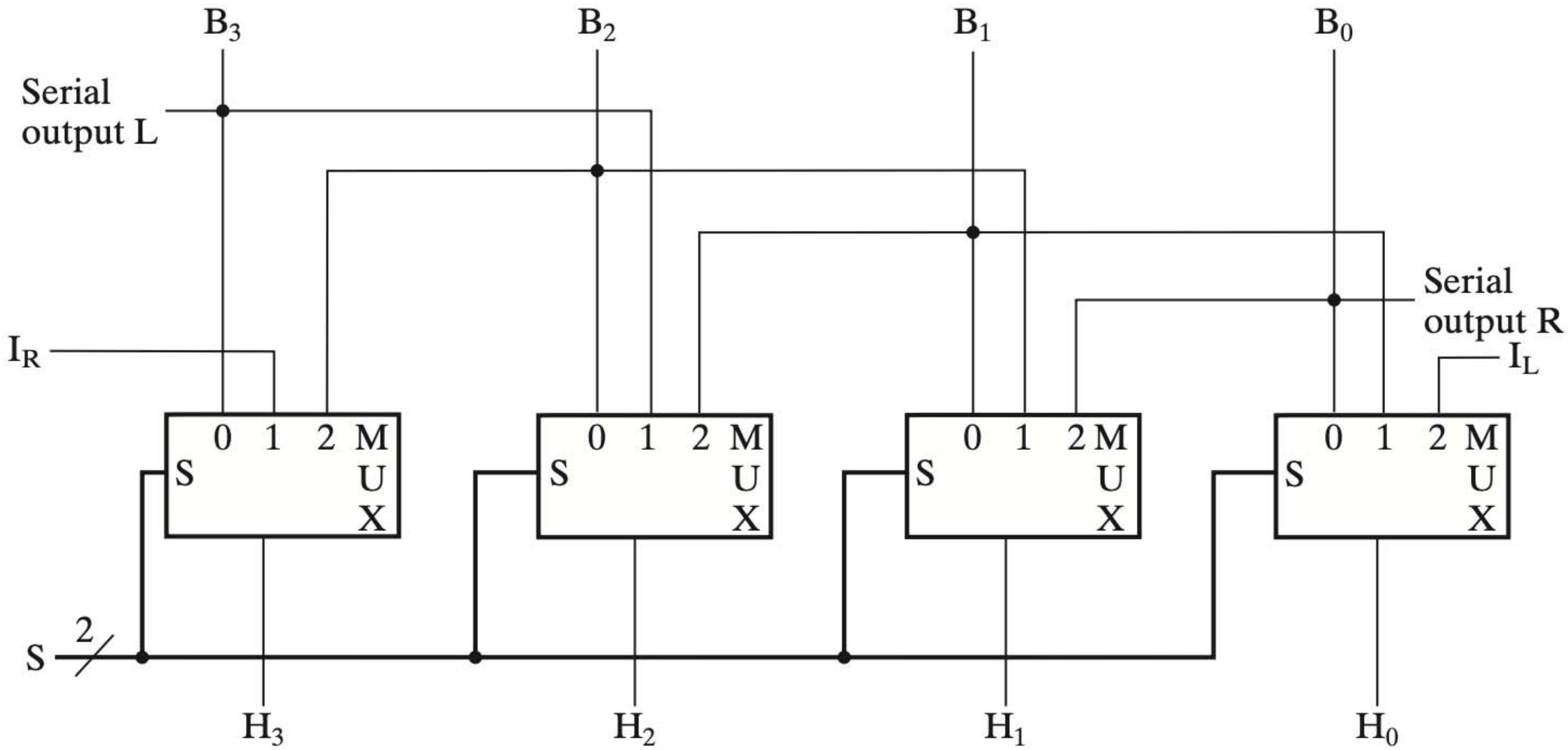
- $A << N = A \times 2^N$
  - E.g., 00001 << 2 = 00100 ($1 \times 2^2 = 4$)
  - E.g., 11101 << 2 = 10100 ($-3 \times 2^2 = -12$)

- $A >>> N = A \div 2^N$
  - E.g., 01000 >>> 2 = 00010 ($8 \div 2^2 = 2$)
  - E.g., 10000 >>> 2 = 11100 ($-16 \div 2^2 = -4$)

Recall the Design of A Shifter

- Bidirectional shift registers with parallel load
  - Data from Bus B can be transferred to the register in parallel and then shifted to the right, the left, or not at all.
  - A clock pulse loads the output of Bus B into the shift register, and a second clock pulse performs the shift.
  - Finally, a third clock pulse transfers the data from the shift register to the selected destination register.
- Alternatively, the transfer from a source register to a destination register can be done using *only one clock pulse* if the shifter is implemented ***as a combinational circuit***
  - Because of the faster operation that results from the use of one clock pulse instead of three, this is the preferred method.
  - In a combinational shifter, the signals propagate through the gates without the need for a clock pulse.
  - Hence, the only clock needed for a shift in the datapath is for loading the data from Bus H into the selected destination register.

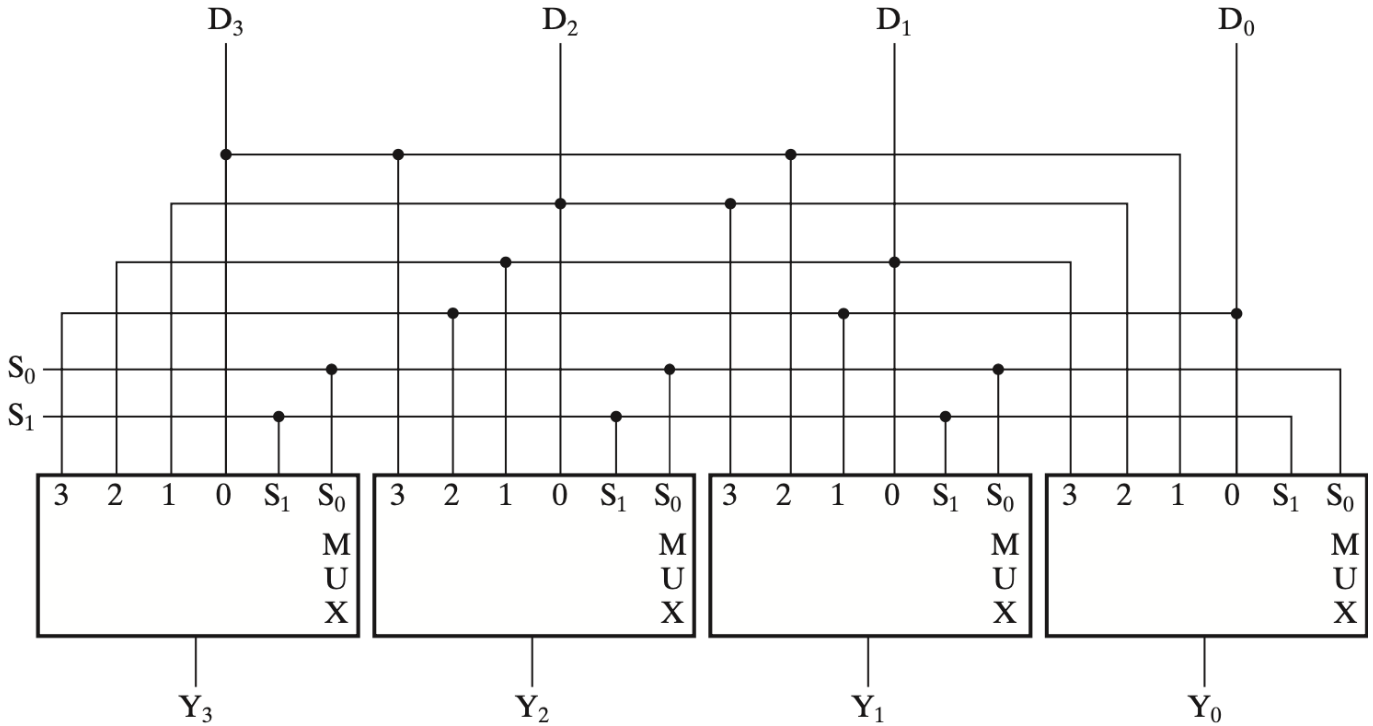
Shifter Design as A Combinational Circuit



4-Bit Barrel Shifter (ROL)

Function Table for 4-Bit Barrel Shifter

Select		Output				
S_1	S_0	Y_3	Y_2	Y_1	Y_0	Operation
0	0	D_3	D_2	D_1	D_0	No rotation
0	1	D_2	D_1	D_0	D_3	Rotate one position
1	0	D_1	D_0	D_3	D_2	Rotate two positions
1	1	D_0	D_3	D_2	D_1	Rotate three positions



Overview

- Addition
- Subtraction
- Arithmetic logic unit (ALU)
- Multiplication
- Division
- Floating number operations

Multiplication

- More complicated than addition
  - A straightforward implementation will involve shifts and adds
- More complex operation can lead to
  - More area (on silicon) and/or
  - More time (multiple cycles or longer clock cycle time)
- Let's begin from a simple, straightforward method

Decimal vs. Binary Multiplication

- Partial products formed by multiplying a single digit of the multiplier with multiplicand
- Shifted partial products summed to form result

Decimal		Binary
230	multiplicand	0101
x 42	multiplier	x 0111
460	partial products	0101
+ 920		0101
9660		0101
		+ 0000
	result	0100011

230 x 42 = 9660

5 x 7 = 35

Unsigned Multiplication: 4-bit X 4-bit

<u>Multiplication</u>

$$\begin{array}{ccccccccl}
 & & & a_3 & a_2 & a_1 & a_0 & & \leftarrow \textit{Multiplicand} \\
 & & \mathrm{X} & b_3 & b_2 & b_1 & b_0 & & \leftarrow \textit{Multiplier} \\
\hline
 & & & a_3b_0 & a_2b_0 & a_1b_0 & a_0b_0 & & \\
 & & a_3b_1 & a_2b_1 & a_1b_1 & a_0b_1 & & & \textit{Partial} \\
 & a_3b_2 & a_2b_2 & a_1b_2 & a_0b_2 & & & & \textit{products} \\
a_3b_3 & a_2b_3 & a_1b_3 & a_0b_3 & & & & & \\
\hline
 & & & \cdots & & a_1b_0+a_0b_1 & a_0b_0 & & \leftarrow \textit{Product}
\end{array}$$

Unsigned Multiplication: M-bit X N-bit

$$A = a_{m-1} \ldots a_1 a_0 \qquad B = b_{n-1} \ldots b_1 b_0$$

$$a = \sum_{i=0}^{m-1} a_i 2^i \qquad b = \sum_{j=0}^{n-1} b_j 2^j$$

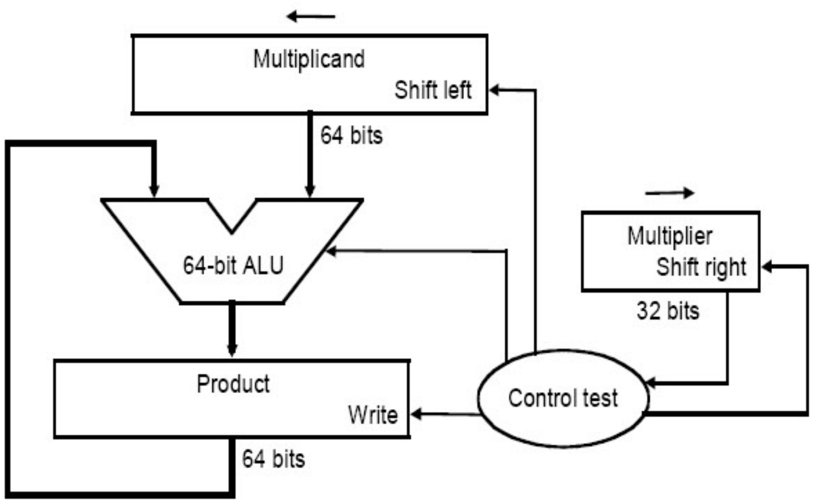
$$p = ab = \left(\sum_{i=0}^{m-1} a_i 2^i\right)\left(\sum_{j=0}^{n-1} b_j 2^j\right) = \sum_{i=0}^{m-1}\sum_{j=0}^{n-1}(a_i b_j)2^{i+j} = \sum_{k=0}^{m+m+1} p_k 2^k$$

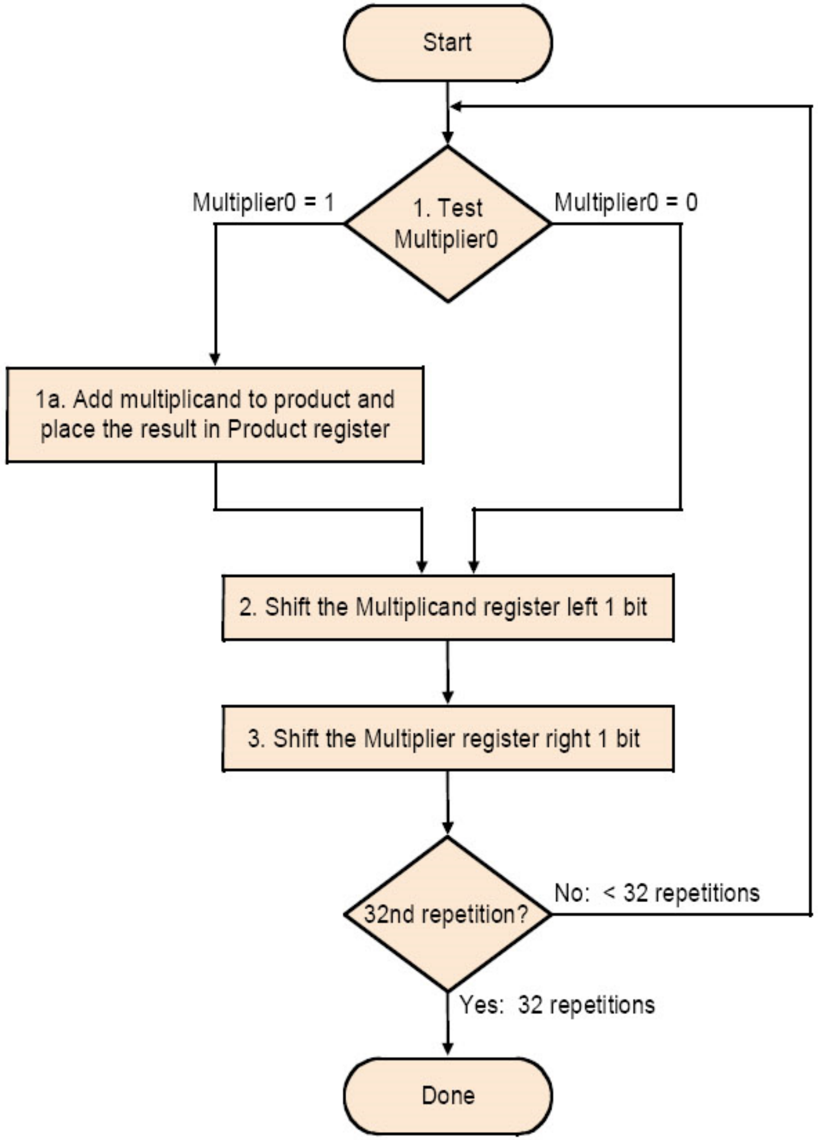
$$\begin{array}{lccccccccc}
 & & & & a_{m-1} & a_{m-2} & \cdots & a_1 & a_0 & = A \\
\times) & & & & & b_{n-1} & \cdots & b_1 & b_0 & = B \\
\hline
 & & & & a_{m-1}b_0 & a_{m-2}b_0 & \cdots & a_1 b_0 & a_0 b_0 & \\
 & & & a_{m-1}b_1 & a_{m-2}b_1 & \cdots & a_1 b_1 & a_0 b_1 & & \\
 & & & \vdots & & & & \vdots & & \\
+) & a_{m-1}b_{n-1} & a_{m-2}b_{n-1} & \cdots & a_1 b_{n-1} & a_0 b_{n-1} & & & & \\
\hline
p_{m+n-1} & p_{m+n-2} & p_{m+n-3} & \cdots & p_{n-1} & \cdots & & p_1 & p_0 & = P
\end{array}$$

Straightforward Algorithm

```
   01010010 (multiplicand)
 x 01101101 (multiplier)
 ----------------------------
        01010010
       00000000
      01010010
     01010010
    00000000
   01010010
  01010010
 00000000
 ----------------------------
 010001011101010
```

Implementation 1





Example of Implementation 1

- Let's do 0010 x 0110 (2 x 6), unsigned

Iteration	Implementation 1			
	Step	Multiplier	Multiplicand	Product
0	initial values	0110	0000 0010	0000 0000
1	1: 0 -> no op	0110	0000 0010	0000 0000
	2: Multiplier shift right/ Multiplicand shift left	011	0000 0100	0000 0000
2	1: 1 -> product = product + multiplicand	011	0000 0100	0000 0100
	2: Multiplier shift right/ Multiplicand shift left	01	0000 1000	0000 0100
3	1: 1 -> product = product + multiplicand	01	0000 1000	0000 1100
	2: Multiplier shift right/ Multiplicand shift left	0	0001 0000	0000 1100
4	1: 0 -> no op	0	0001 0000	0000 1100
	2: Multiplier shift right/Multiplicand shift left		0010 0000	

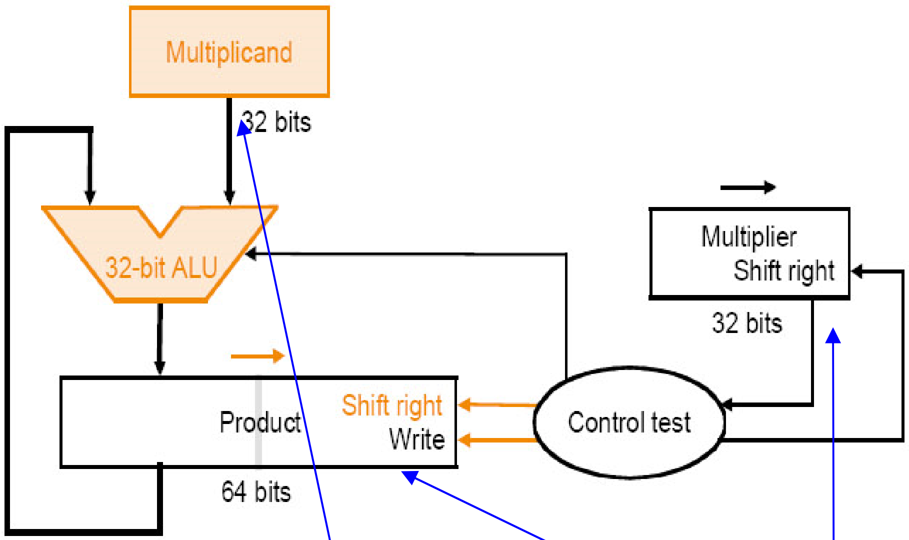
Drawbacks

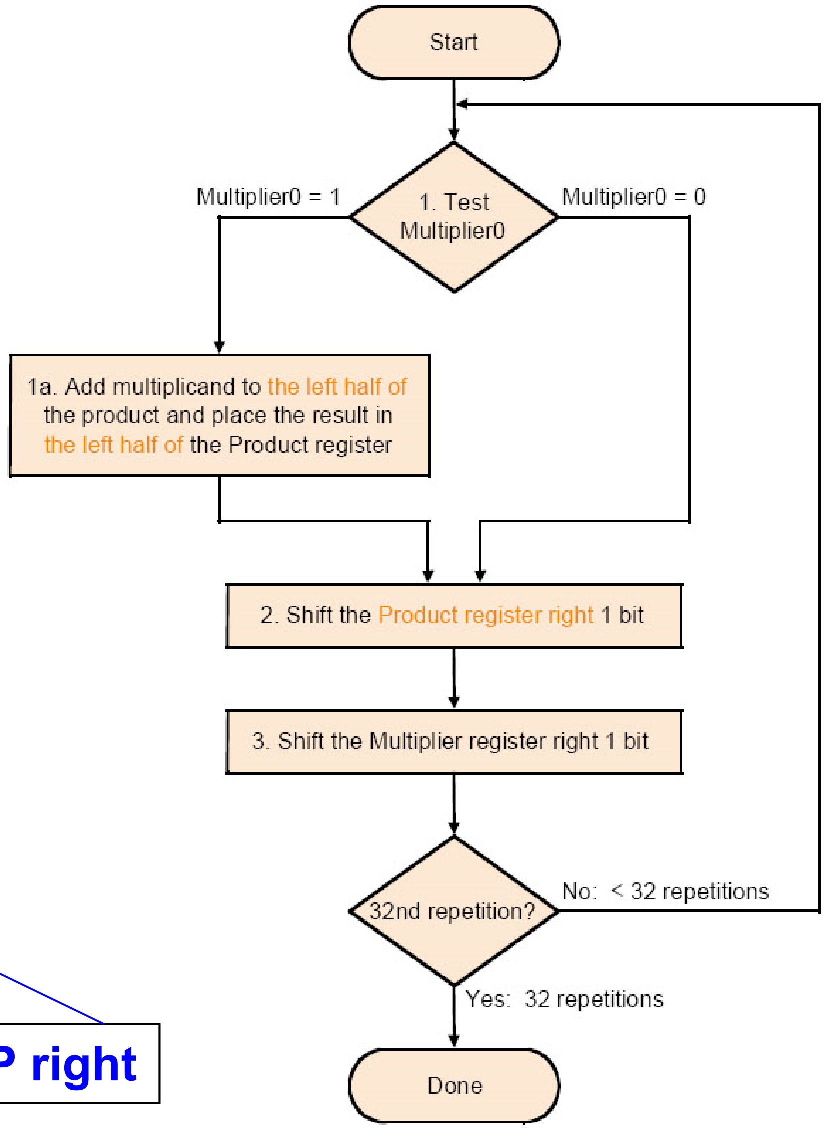
- The ALU is twice as wide as necessary
- The multiplicand register takes twice as many bits as needed
- The product register won't need 2n bits till the last step
  - Being filled
- The multiplier register is being emptied during the process

Improved Algorithm

```
        01010010 (multiplicand)
      x 01101101 (multiplier)
        01010010
       00000000
      01010010
     01010010
    00000000
   01010010
  01010010
 00000000
 010001011101010
```

Implementation 2





Multiplicand stationary - Multiplier right - PP right

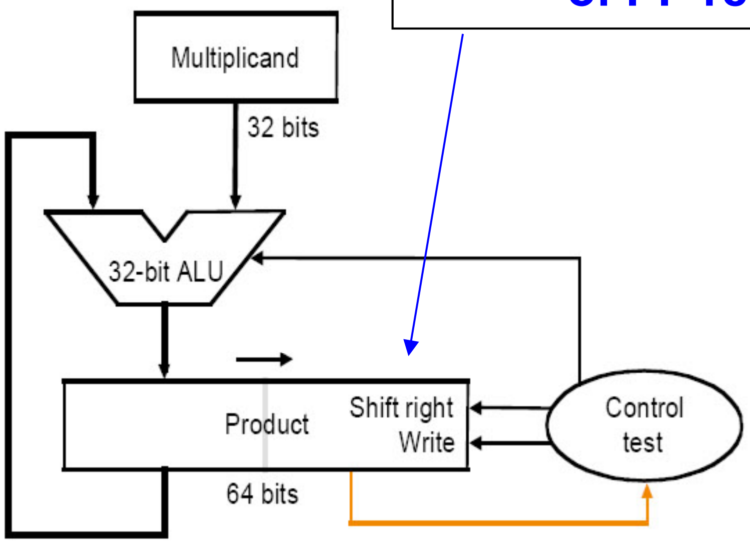
Example of Implementation 2

- Let's do 0010 x 0110 (2 x 6), unsigned

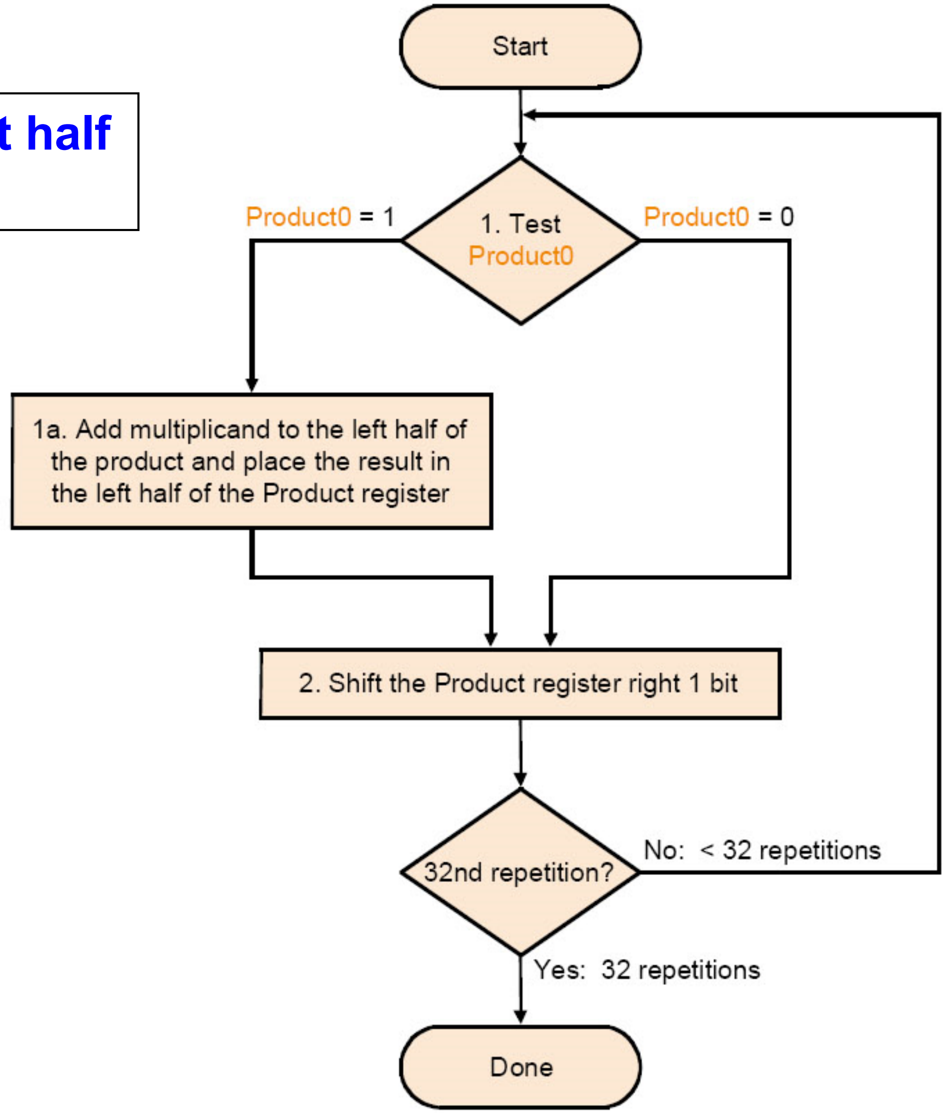
Iteration	Implementation 2			
	Step	Multiplier	Multiplicand	Product
0	initial values	0110	0010	0000 $_{\times\times\times\times}$
1	1: 0 -> no op	0110	0010	0000 $_{\times\times\times\times}$
	2: Multiplier shift right/ Product shift right	$_{\times}$011	0010	0000 0$_{\times\times\times}$
2	1: 1 -> product = product + multiplicand	$_{\times}$011	0010	0010 0$_{\times\times\times}$
	2: Multiplier shift right/ Product shift right	$_{\times\times}$01	0010	0001 00$_{\times\times}$
3	1: 1 -> product = product + multiplicand	$_{\times\times}$01	0010	0011 00$_{\times\times}$
	2: Multiplier shift right/ Product shift right	$_{\times\times\times}$0	0010	0001 100$_{\times}$
4	1: 0 -> no op	$_{\times\times\times}$0	0010	0001 100$_{\times}$
	2: Multiplier shift right/ Product shift right	$_{\times\times\times\times}$	0010	0000 1100

Implementation 3

Multiplier on right half of PP reg



Multiplicand stationary - Multiplier right - PP right



Example of Implementation 3

- Let’s do 0010 x 0110 (2 x 6), unsigned

Iteration	Implementation 3			
	Step	Multiplier	Multiplicand	Product|Multiplier
0	initial values	0110	0010	0000 0110
1	1: 0 -> no op	0110	0010	0000 0110
	2: Multiplier shift right/ Product shift right	x011	0010	0000 0011
2	1: 1 -> product = product + multiplicand	x011	0010	0010 0011
	2: Multiplier shift right/ Product shift right	xx01	0010	0001 0001
3	1: 1 -> product = product + multiplicand	xx01	0010	0011 0001
	2: Multiplier shift right/ Product shift right	xx00	0010	0001 1000
4	1: 0 -> no op	xxx0	0010	0001 1000
	2: Multiplier shift right/ Product shift right	xxxx	0010	0000 1100

Signed Multiplication

- Recall
  - For p = a x b, if a < 0 or b < 0, then p < 0
  - If a < 0 and b < 0, then p > 0
  - Hence sign(p) = sign(a) xor sign(b)
- Hence
  - Convert multiplier, multiplicand to positive number with (n-1) bits
  - Multiply positive numbers
  - Compute sign, convert product accordingly
  - Or, Perform sign-extension on shifts for prev. design

Example of Signed Multiplication

- Note:
  - Sign extension of partial product
  - If most significant bit of multiplier is 1, then subtract

$00111_2 * 0111_2$	$11001_2 * 1001_2$	$00111_2 * 1001_2$	$11001_2 * 0111_2$
00000 0111	00000 1001	00000 1001	00000 0111
+ 00111 0111	+ 11001 1001	+ 00111 1001	+ 11001 0111
→ 00011 1011	→ 11100 1100	→ 00011 1100	→ 11100 1011
+ 01010 1011	→ 11110 0110	→ 00001 1110	+ 10101 1011
→ 00101 0101	→ 11111 0011	→ 00000 1111	→ 11010 1101
+ 01100 0101	subtract	subtract	+ 10011 1101
→ 00110 0010	00110 0011	11001 1111	→ 11001 1110
→ 00011 0001	→ 00011 0001	→ 11100 1111	→ 11100 1111

Booth's Algorithm Principle

- Assumes： $Z=y\times10111100$

$Z=y(10000000+111100+100-100)$

$=y(1\times2^7+1000000-100)$

$=y(1\times2^7+1\times2^6-2^2)$

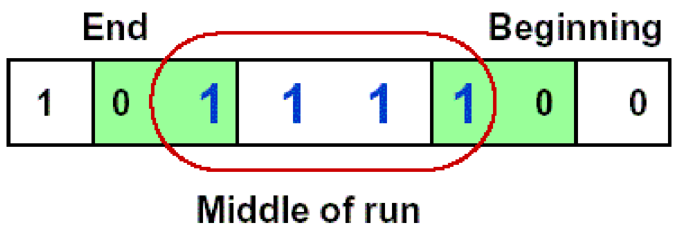
$=y(1\times2^7+1\times2^6+0\times2^5+0\times2^4+0\times2^3+0\times2^2+0\times2^1+0\times2^0-1\times2^2)$

$=y(1\times2^7+1\times2^6+0\times2^5+0\times2^4+0\times2^3+0\times2^2-1\times2^2+0\times2^1+0\times2^0)$

$=y\times2^7+y\times1\times2^6+0\times2^5+0\times2^4+0\times2^3+0\times2^2-y\times2^2$
$+0\times2^1+0\times2^0)$

Booth's Algorithm

- Idea: If you have a sequence of '1's
  - subtract at first '1' in multiplier
  - shift for the sequence of '1's
  - add where prior step had last '1'



- Result:
  - Possibly less additions and more shifts
  - Faster, if shifts are faster than additions

Booth’s Algorithm

Current bit	Bit to right	Explanation	Example	Operation
1	0	Begins run of ‘1’	0000111100	Subtract
1	1	Middle of run of ‘1’	00001111000	Nothing
0	1	End of a run of ‘1’	00001111000	Add
0	0	Middle of a run of ‘0’	00001111000	Nothing

Example 1 of Booth’s Algorithm

- Let’s do 0010 x 1101 (2 x -3)

Iteration	Implementation 3		
	Step	Multiplicand	Product
0	initial values	0010	0000 1101 0
1	10 -> product = product – multiplicand	0010	1110 1101 0
	shift right		1111 0110 1
2	01 -> product = product + multiplicand	0010	0001 0110 1
	shift right		0000 1011 0
3	10 -> product = product – multiplicand	0010	1110 1011 0
	shift right		1111 0101 1
4	11 -> no op	0010	1111 0101 1
	shift right		1111 1010 1

Example 2 of Booth's Algorithm

- Negative multiplicand:
  - -6 x 6 = -36
  - 1010 x 0110, 0110 in Booth's encoding is +0-0
  - Hence:

1111 1010	x 0	0000 0000
1111 0100	x –1	0000 1100
1110 1000	x 0	0000 0000
1101 0000	x +1	1101 0000
	Final Sum:	1101 1100 (-36)

Example 3 of Booth's Algorithm

- Negative multiplier:
  - -6 x -2 = 12
  - 1010 x 1110, 1110 in Booth's encoding is 00-0
  - Hence:

1111 1010	x 0	0000 0000
1111 0100	x –1	0000 1100
1110 1000	x 0	0000 0000
1101 0000	x 0	0000 0000
	Final Sum:	0000 1100 (12)

Summary

```
        0 0 1 0 1 1
        0 1 0 0 1 1
        -----------
        0 0 1 0 1 1
      0 0 1 0 1 1
    0 0 0 0 0 0
  0 0 0 0 0 0
0 0 1 0 1 1
-------------------
0 0 1 1 0 1 0 0 0 1
```

Benefit: Reducing the number of partial products

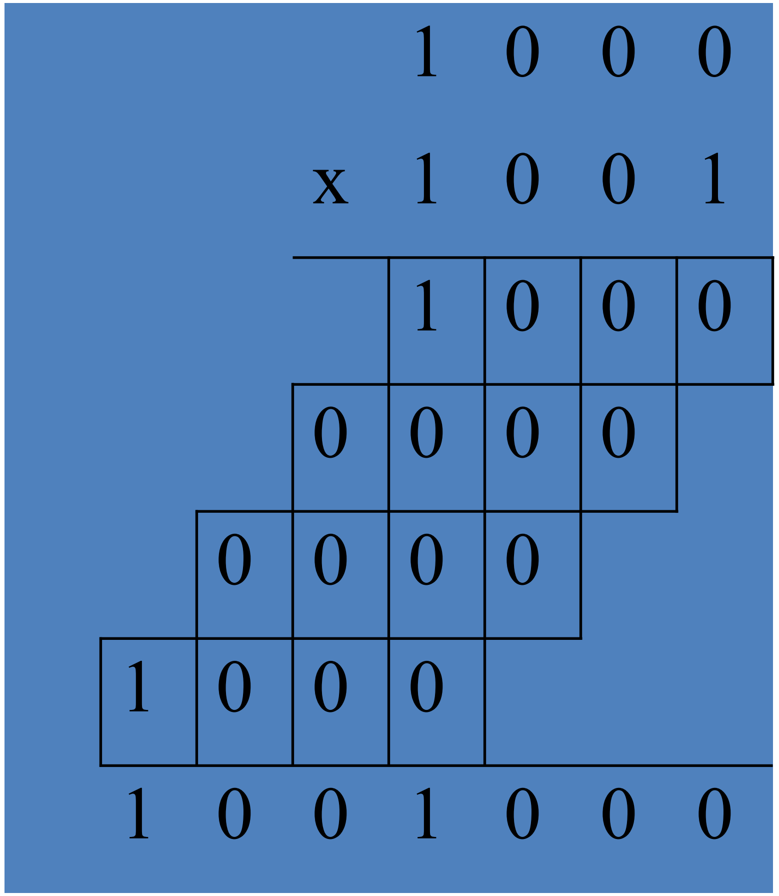
Take away: Booth encoding

Exercise

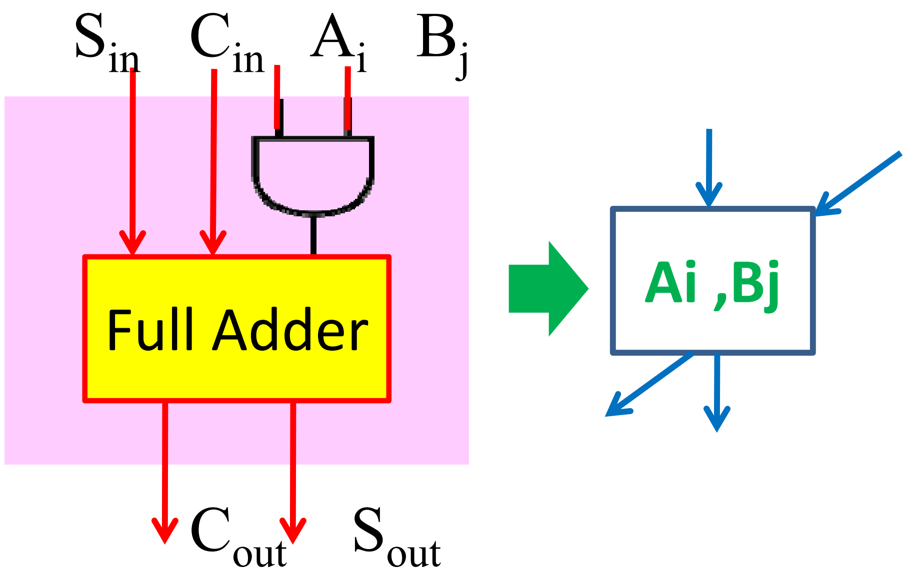
Unsigned Multiplication: $x=5$, $y=3$

Signed Multiplication: $x=5$, $y=-7$

Yet Another Option: Array Multiplier

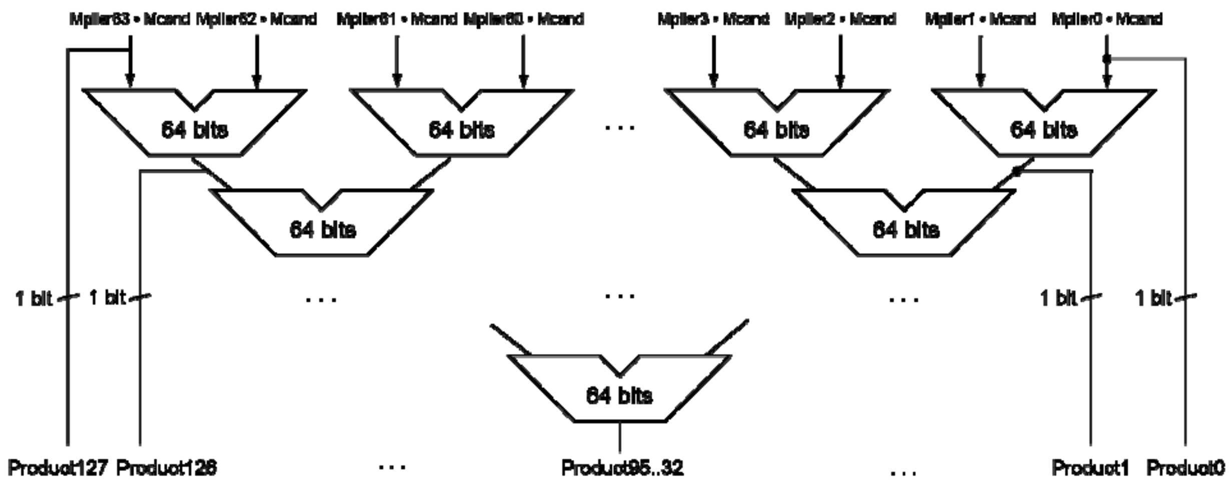


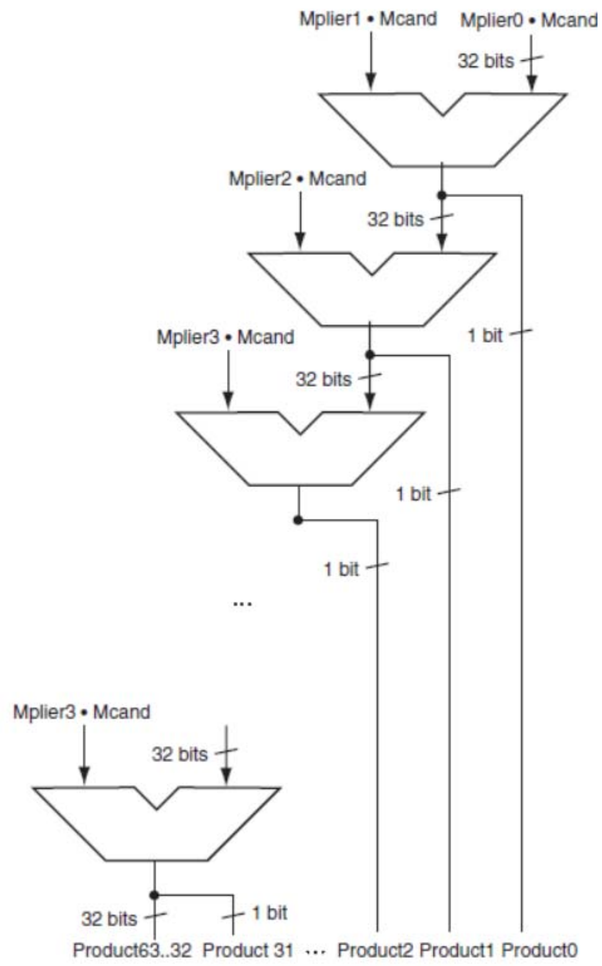
Adding all partial products simultaneously using an array of basic cells



Faster Multiplication

- Uses multiple adders
  - Cost/performance tradeoff





- Can be pipelined
  - Several multiplication performed in parallel

Overview

- Addition
- Subtraction
- Arithmetic logic unit (ALU)
- Multiplication
- Division
- Floating number operations

Unsigned Division

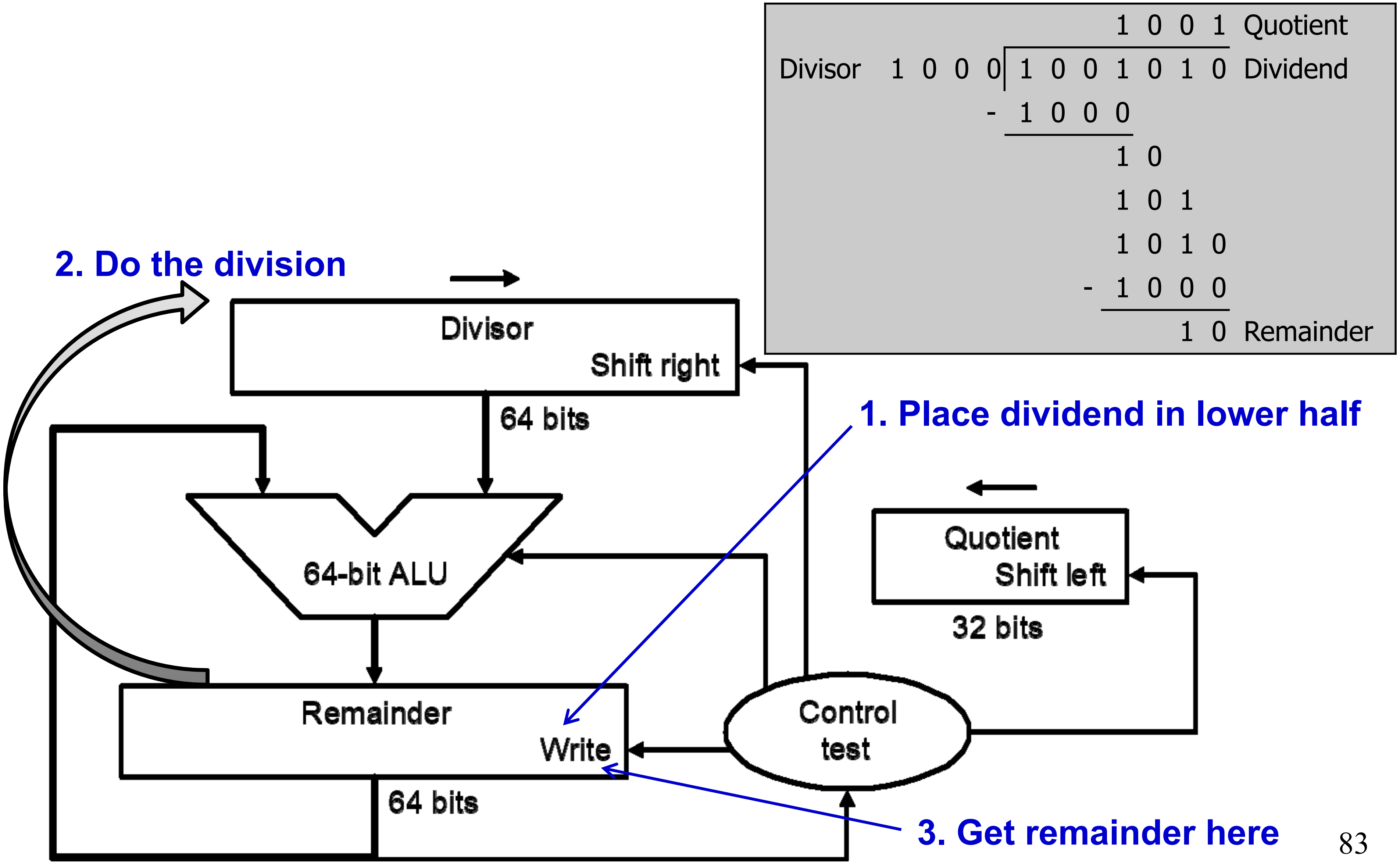
- Again, back to 3rd grade ($74 \div 8 = 9$ rem 2)

								1	0	0	1	Quotient
Divisor	1	0	0	0	1	0	0	1	0	1	0	Dividend
				-	1	0	0	0				
								1	0			
								1	0	1		
								1	0	1	0	
							-	1	0	0	0	
										1	0	Remainder

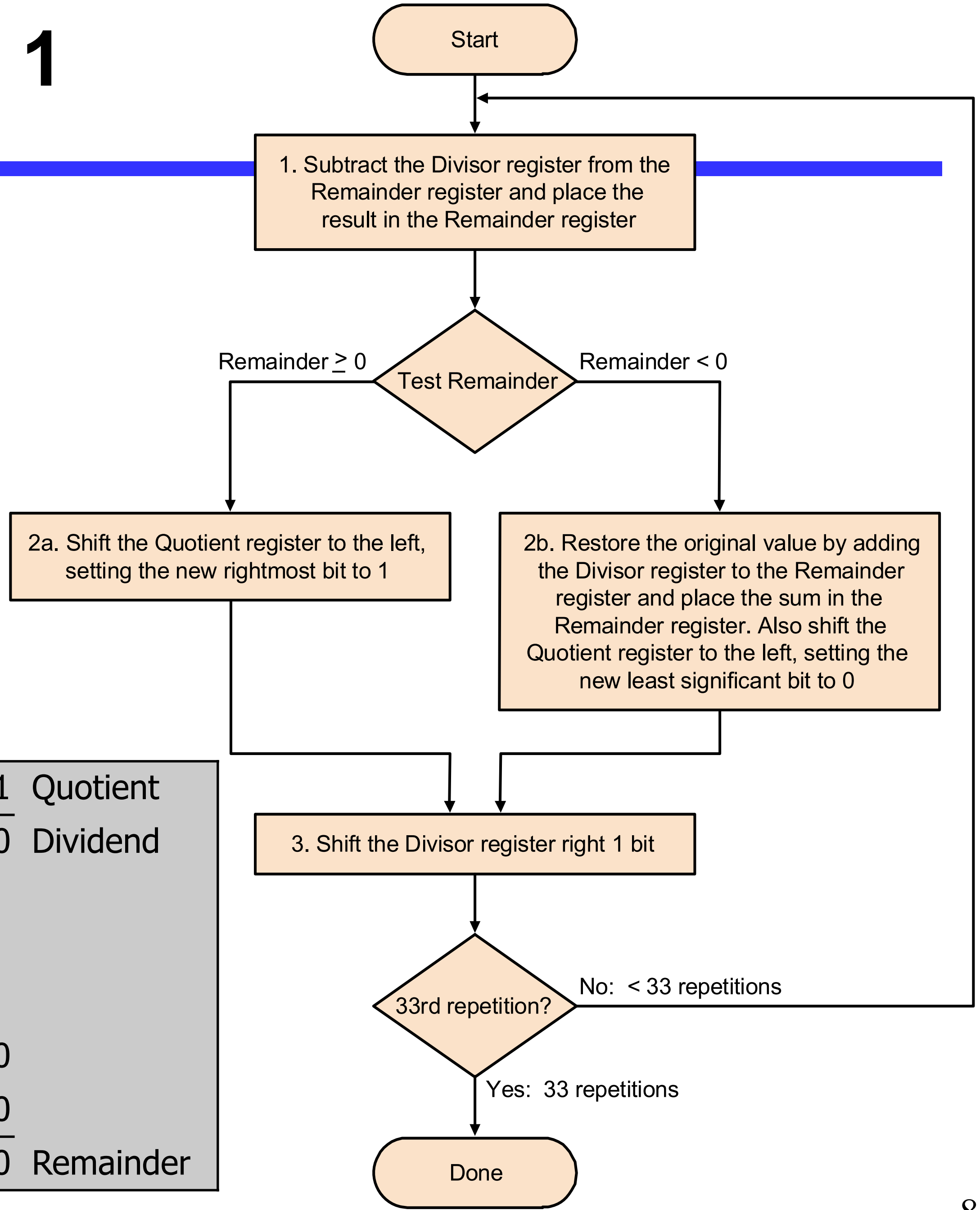
Unsigned Division

- How does hardware know if division fits?
  - Condition: if remainder ≥ divisor
  - Use subtraction: (remainder – divisor) ≥ 0
- OK, so if it fits, what do we do?
  - $\text{Remainder}_{n+1} = \text{Remainder}_n - \text{divisor}$
- What if it doesn't fit?
  - Have to restore original remainder
- Called **restoring division**

Implementation 1



Implementation 1




```
                     1 0 0 1   Quotient
Divisor  1 0 0 0 | 1 0 0 1 0 1 0   Dividend
                - 1 0 0 0
                        1 0
                        1 0 1
                        1 0 1 0
                      - 1 0 0 0
                            1 0   Remainder
```

Example of Implementation 1

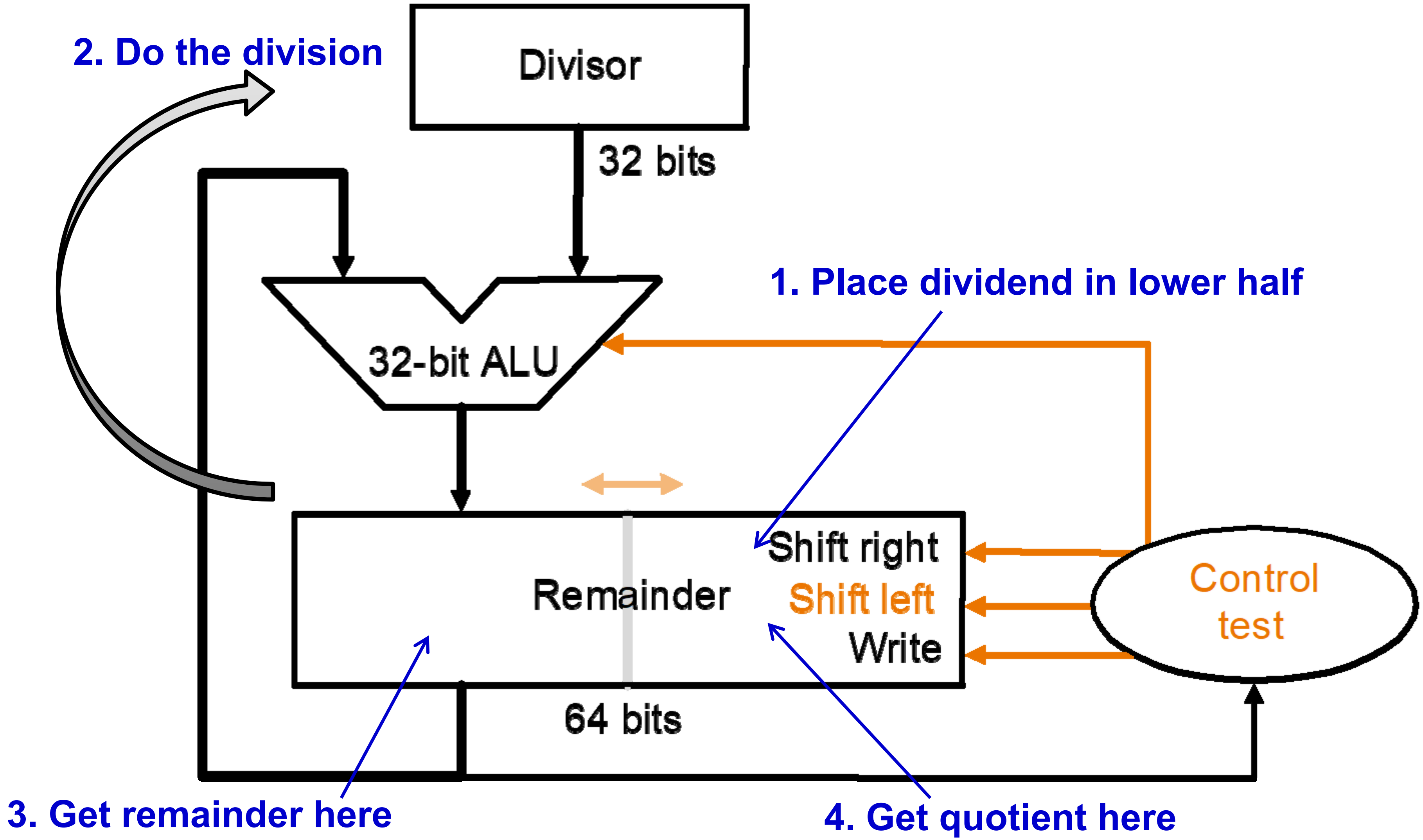
- Let's do 7 ÷ 2

Iteration	Step	Quotient	Divisor	Remainder
0	Initial values	0000	0010 0000	0000 0111
1	1: Rem=Rem-Div	0000	0010 0000	1110 0111
	2b: Rem<0=>+Div, sll Q, Q_0=0	0000	0010 0000	0000 0111
	3: Shift Div right	0000	0001 0000	0000 0111
2	1: Rem=Rem-Div	0000	0001 0000	1111 0111
	2b: Rem<0=>+Div, sll Q, Q_0=0	0000	0001 0000	0000 0111
	3: Shift Div right	0000	0000 1000	0000 0111
3	1: Rem=Rem-Div	0000	0000 1000	1111 1111
	2b: Rem<0=>+Div, sll Q, Q_0=0	0000	0000 1000	0000 0111
	3: Shift Div right	0000	0000 0100	0000 0111
4	1: Rem=Rem-Div	0000	0000 0100	0000 0011
	2a: Rem≥0=> sll Q, Q_0=1	0001	0000 0100	0000 0011
	3: Shift Div right	0001	0000 0010	0000 0011
5	1: Rem=Rem-Div	0001	0000 0010	0000 0001
	2a: Rem≥0=> sll Q, Q_0=1	0011	0000 0010	0000 0001
	3: Shift Div right	0011	0000 0001	0000 0001

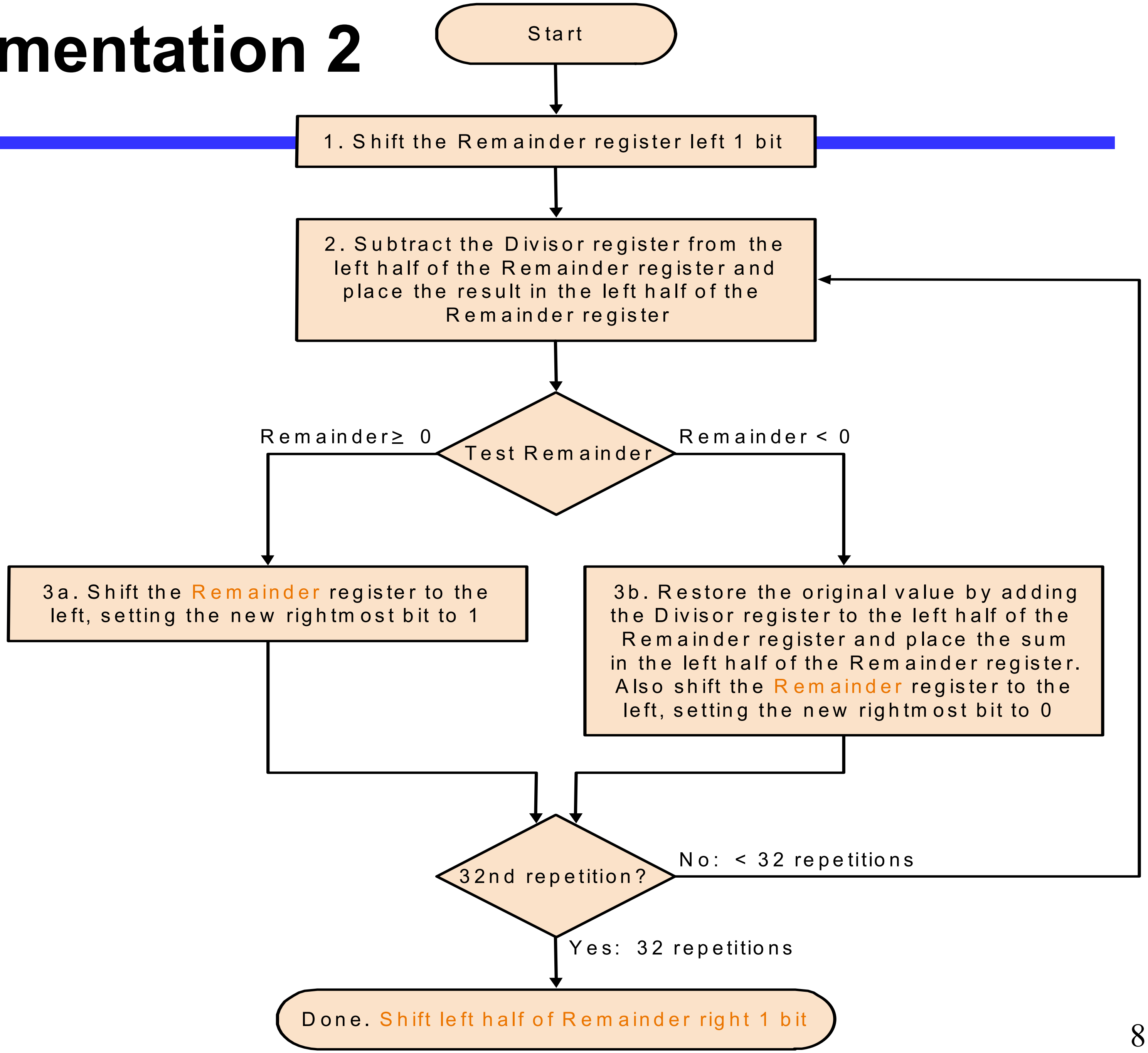
Division Improvements

- Skip first subtract
  - Can’t shift ‘1’ into quotient anyway
  - Hence shift first, then subtract
    - Undo extra shift at end
- Hardware similar to multiplier
  - Can store quotient in remainder register
  - Only need 32-bit ALU
    - Shift remainder left vs. divisor right

Implementation 2



Implementation 2



Example 7/2 for Restoring Division

- Well known numbers: 0000 0111/0010

iteration	step	Divisor	Remainder
0	Initial Values	0010	0000 0111
	Shift Rem left 1	0010	**0000 1110**
1	1.Rem=Rem-Div	0010	1110 1110
	2b: Rem<0 →+Div,sll R,R_0=0	0010	**0001 *1100***
2	1.Rem=Rem-Div	0010	1111 1100
	2b: Rem<0 →+Div,sll R,R_0=0	0010	**0011 *1000***
3	1.Rem=Rem-Div	0010	0001 1000
	2a: Rem>0 →sll R,R_0=1	0010	**0011 *0001***
4	1.Rem=Rem-Div	0010	0001 0001
	2a: Rem>0 →sll R,R_0=1	0010	**0010 *0011***
	Shift left half of Rem right 1		**0001 *0011***

Further Improvements

- Division still takes:
  - 2 ALU cycles per bit position
    - 1 to check for divisibility (subtract)
    - One to restore (if needed)
- Can reduce to 1 cycle per bit
  - Called **non-restoring division**
  - Avoids restore of remainder when test fails

Non-Restoring Division

- Consider remainder to be restored:

  $$\mathbf{R_i = 2 \times R_{i-1} - d < 0}$$

  - Since R_i is negative, we must restore it, right?
  - Well, maybe not. Consider next step i+1:

  $$\mathbf{R_{i+1} = 2 \times (R_i) - d = 2 \times (R_i - d) + d}$$

- Hence, we can compute R_{i+1} by not restoring R_i, and adding d instead of subtracting d
  - Same value for R_{i+1} results
- Throughput of 1 bit per cycle

Non-Restoring Division

Iteration	Step	Divisor	Remainder
0	Initial values	0010	0000 0111
	Shift rem left 1	0010	0000 1110
1	2: Rem = Rem - Div	0010	1110 1110
	3b: Rem < 0 (add next), sll 0	0010	1101 1100
2	2: Rem = Rem + Div	0010	1111 1100
	3b: Rem < 0 (add next), sll 0	0010	1111 1000
3	2: Rem = Rem + Div	0010	0001 1000
	3a: Rem > 0 (sub next), sll 1	0010	0011 0001
4	Rem = Rem – Div	0010	0001 0001
	Rem > 0 (sub next), sll 1	0010	0010 0011
	Shift Rem right by 1	0010	0001 0011

What About Signed Division?

- The simplest solution is to remember the signs of the divisor and dividend and then negate the quotient if the signs disagree.
- However, $(-7) \div 2 = ?$
  - $(+7) \div (-2) = ?$
  - $(-7) \div (-2) = ?$

- The correctly signed division algorithm *negates the quotient if the signs of the operands are opposite and makes the sign of the nonzero remainder match the dividend*.

Overview

- Addition
- Subtraction
- Arithmetic logic unit (ALU)
- Multiplication
- Division
- Floating number operations

Floating point addition

- Example in decimal： system precision 4 digits

  What is $9.999 \cdot 10^{1} + 1.610 \cdot 10^{-1}$?
- Aligning the two numbers

  $9.999 \cdot 10^{1}$

  $0.016\textcolor{red}{10} \cdot 10^{1} \rightarrow 0.016 \cdot 10^{1}$ Truncation
- Addition

$$\begin{array}{rl} 9.999 & \cdot 10^{1} \\ +\ 0.016 & \cdot 10^{1} \\ \hline 10.015 & \cdot 10^{1} \end{array}$$

- Normalization

  $1.0015 \cdot 10^{2}$
- Rounding

  $1.002 \cdot 10^{2}$

Floating point addition steps

- Alignment
- The proper digits have to be added
- Addition of significands
- Normalization of the result
- Rounding

Example y=0.5+(-0.4375) in binary

- $0.5_{10} = 1.000_2 \times 2^{-1}$
- $-0.4375_2 = -1.110_2 \times 2^{-2}$
- Step1:The fraction with lesser exponent is shifted right until matches

$$-1.110_2 \times 2^{-2} \rightarrow -0.111_2 \times 2^{-1}$$

- Step2: Add the significands

$$\begin{array}{rl} & \mathbf{1.000_2 \times 2^{-1}} \\ +) & \mathbf{-\ 0.111_2 \times 2^{-1}} \\ \hline & \mathbf{0.001_2 \times 2^{-1}} \end{array}$$

- Step3: Normalize the sum and checking for overflow or underflow

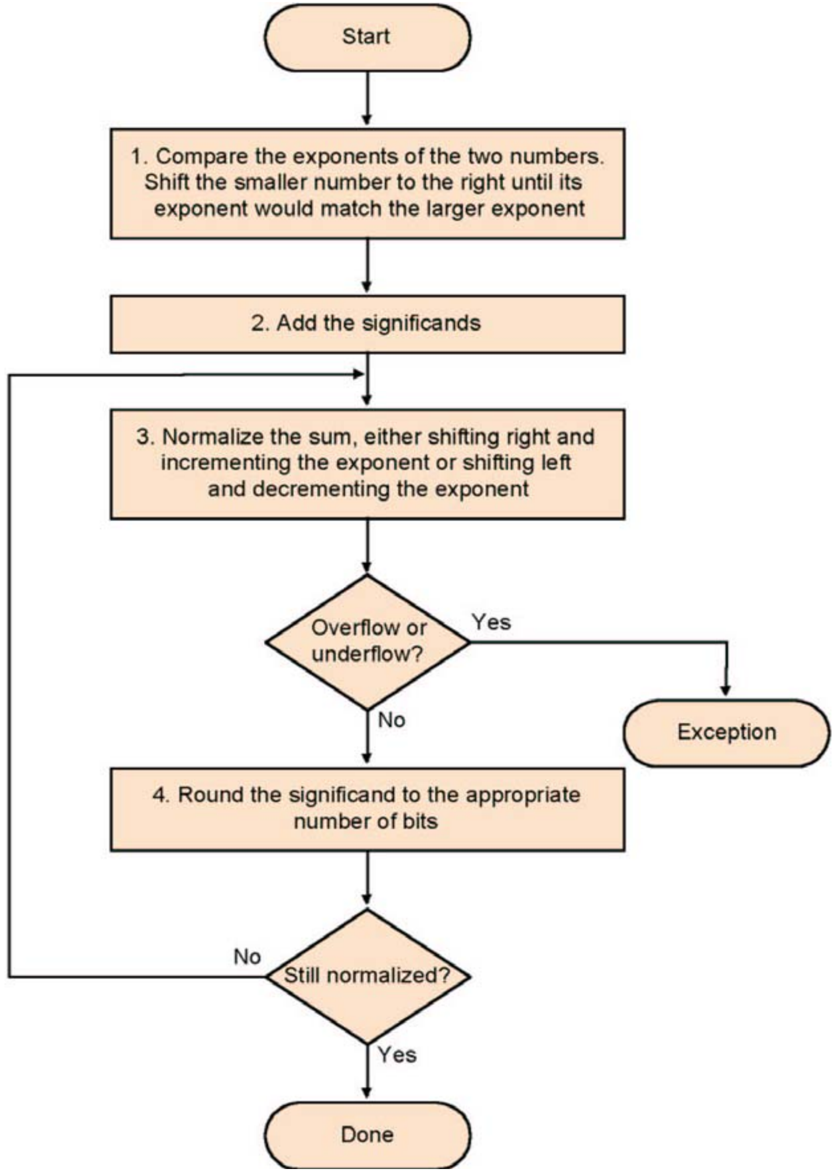
$$\mathbf{0.001_2 \times 2^{-1} \rightarrow 0.010_2 \times 2^{-2} \rightarrow 0.100_2 \times 2^{-3} \rightarrow 1.000_2 \times 2^{-4}}$$

- Step4: Round the sum

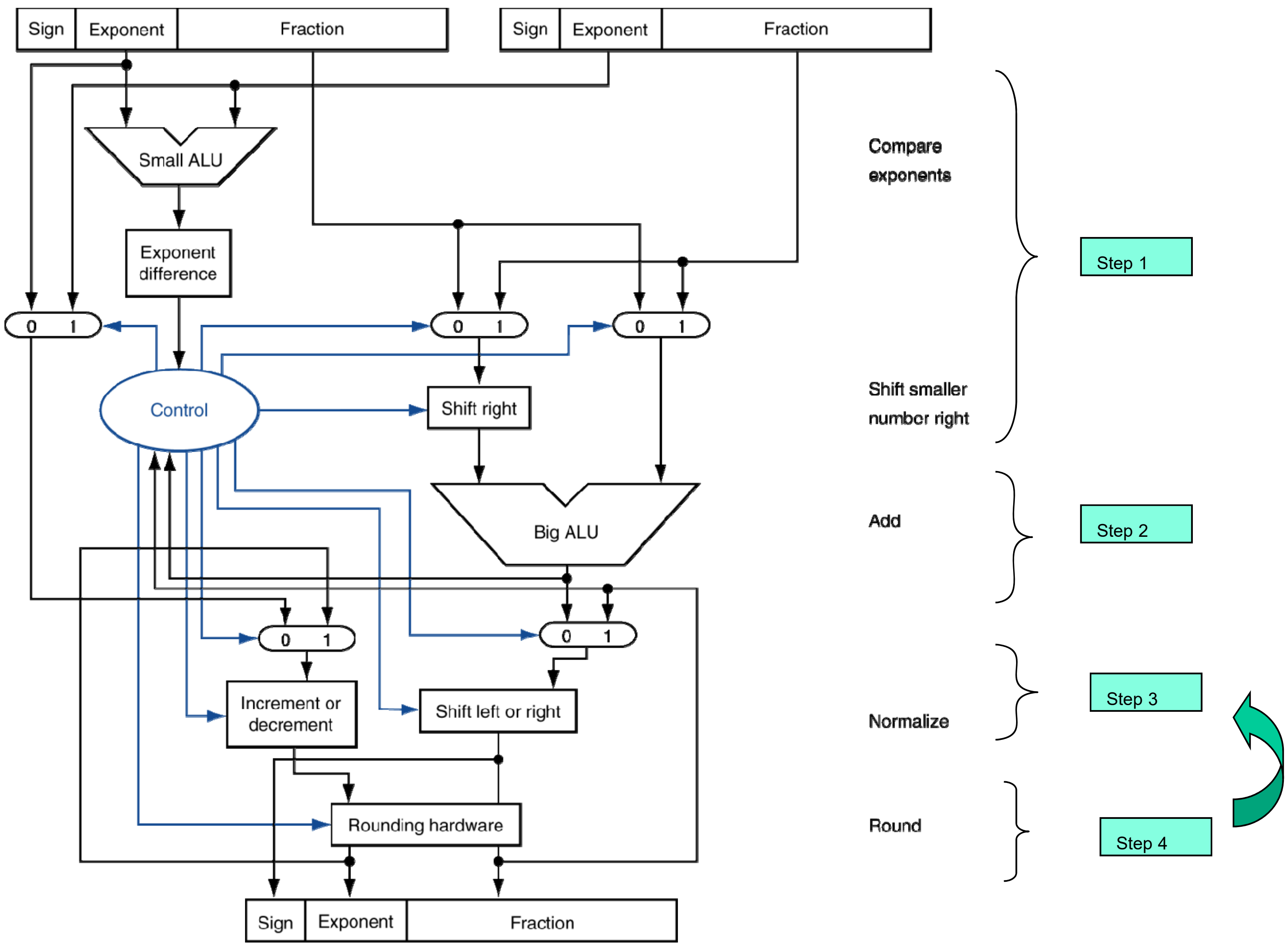
$$\mathbf{1.000_2 \times 2^{-4} = 0.0625}_{10}$$

Algorithm

- Normalize Significands
- Add Significands
- Normalize the sum
- Over/underflow
- Rounding
- Normalization



FP Adder Hardware



FP Adder Hardware

- Much more complex than integer adder
- Doing it in one clock cycle would take too long
  - Much longer than integer operations
  - Slower clock would penalize all instructions
- FP adder usually takes several cycles
  - Can be pipelined

Floating-Point Multiplication

- Consider a 4-digit decimal example
  - $1.110 \times 10^{10} \times 9.200 \times 10^{-5}$
- 1. Add exponents
  - For biased exponents, subtract bias from sum
  - New exponent $= 10 + -5 = 5$
- 2. Multiply significands
  - $1.110 \times 9.200 = 10.212 \;\Rightarrow\; 10.212 \times 10^5$
- 3. Normalize result & check for over/underflow
  - 1.0212×10^6
- 4. Round and renormalize if necessary
  - 1.021×10^6
- 5. Determine sign of result from signs of operands
  - $+1.021 \times 10^6$

Floating Point Multiplication

- Composition of number from different parts
  $\rightarrow$ separate handling

$$(s1 \bullet 2^{e1}) \bullet (s2 \bullet 2^{e2}) = (s1 \bullet s2) \bullet 2^{e1+e2}$$

- Example

  1 10000010 000 0000 0000 0000 0000 0000 = -1×2^3

  0 10000011 000 0000 0000 0000 0000 0000 = 1×2^4

- Both significands are 1 $\rightarrow$ product = 1 $\rightarrow$Sign=1

- Add the exponents, bias = 127

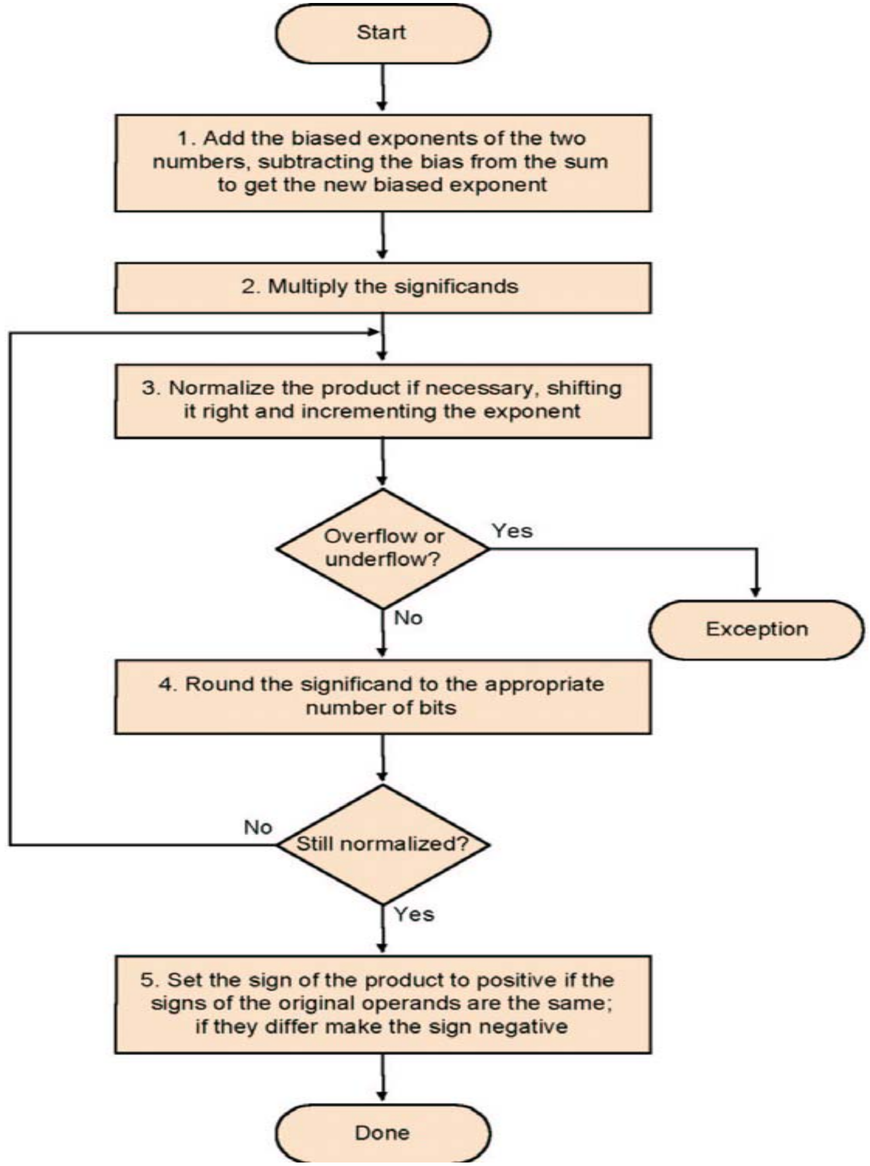
$$\begin{array}{r} 10000010 \\ \underline{+10000011} \\ 110000101 \end{array}$$

  Correction: 110000101-01111111=10000110=134=127+3+4

- The result: 1 10000110 000 0000 0000 0000 0000 0000 = -1×2^7

Multiplication

- Add exponents
- Multiply the significands
- Normalize
- Over- underflow
- Rounding
- Sign

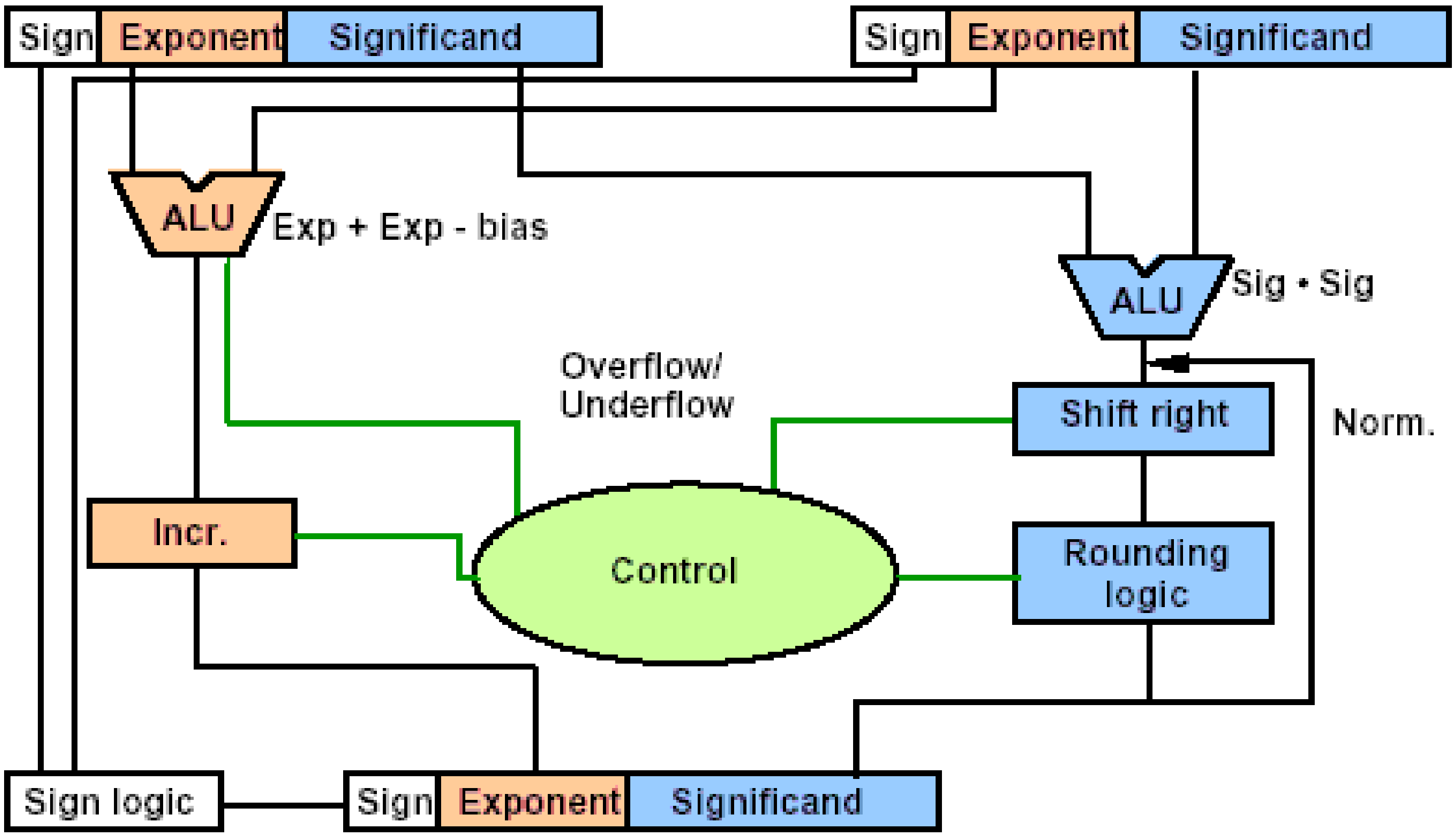


multiplying the numbers 0.5_{ten} and -0.4375_{ten} $\rightarrow 1.000_{two} \times 2^{-1}$ by $-1.110_{two} \times 2^{-2}$

- Step1:Adding the exponents without bias or using the biased
  - $-1 + (-2) = -3$
  - $(-1 + 127) + (-2 + 127) - 127 = (-1 - 2) + (127 + 127 - 127)$
    $= -3 + 127 = 124$
- Step 2. Multiplying the significands
  - $1.110000_{two} x2^{-3}$
- Step 3. normalize
  - $127 \geq -3 \geq -126$, no overflow or underflow.
- Step 4. Rounding
  - $1.110_{two} x2^{-3}$
- Step 5. sign
  - $-1.110_{two} x2^{-3} = -0.21875_{ten}$

$$
\begin{array}{rr}
 & 1.000_{two} \\
\times & 1.110_{two} \\
\hline
 & 0000 \\
 & 1000 \\
 & 1000 \\
 & 1000 \\
\hline
 & 1110000_{two}
\end{array}
$$

FP Multiplier Hardware



FP Arithmetic Hardware

- FP multiplier is of similar complexity to FP adder
  - But uses a multiplier for significands instead of an adder
- FP arithmetic hardware usually does
  - Addition, subtraction, multiplication, division, reciprocal, square-root
  - FP $\leftrightarrow$ integer conversion
- Operations usually takes several cycles
  - Can be pipelined

Division-- Brief

- Subtraction of exponents
- Division of the significands
- Normalization
- Rounding
- Sign

END

Half Subtractor

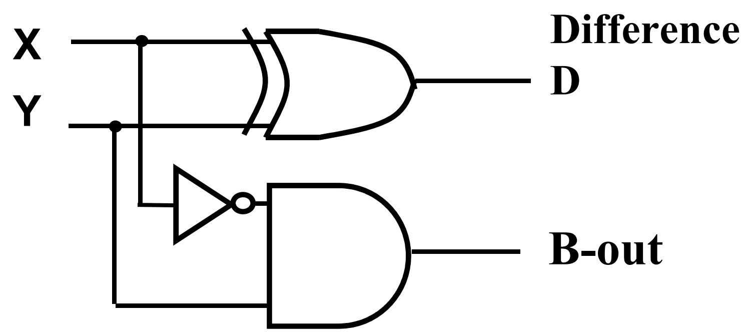
- Subtracting a single-bit binary value Y from anther X (i.e., X -Y) produces a difference bit D and a borrow out bit B-out.
- This operation is called half subtraction and the circuit to realize it is called a half subtractor.

Half Subtractor Truth Table

Inputs		Outputs	
X	**Y**	**D**	**B-out**
0	0	0	0
0	1	1	1
1	0	1	0
1	1	0	0

$$D = X'Y + XY' = X \oplus Y$$

$$\text{B-out} = X'Y$$



Full Subtractor

- Subtracting two single-bit binary values, Y, B-in from a single-bit value X produces a difference bit D and a borrow out B-out bit. This is called full subtraction.

Full Subtractor Truth Table

Inputs			Outputs	
X	**Y**	**B-in**	**D**	**B-out**
0	0	0	0	0
0	0	1	1	1
0	1	0	1	1
0	1	1	0	1
1	0	0	1	0
1	0	1	0	0
1	1	0	0	0
1	1	1	1	1

$$D(X,Y,B_{in}) = \Sigma\,(1,2,4,7)$$
$$B_{out}(X,Y,B_{in}) = \Sigma\,(1,2,3,7)$$

Difference D

B_{in} \ XY	00	01	11	10
0	0	2 1	6	4 1
1	1 1	3	7 1	5

X over columns 11, 10; Y over columns 01, 11; B_{in} beside row 1.

$$D = X'Y'(B_{in}) + XY'(B_{in})' + XY'(B_{in})' + XY(B_{in})$$
$$D = X \oplus Y \oplus (B_{in})$$

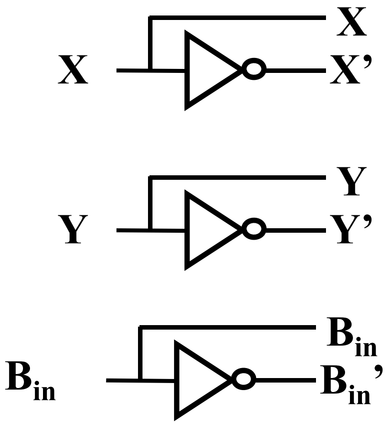
Borrow B_{out}

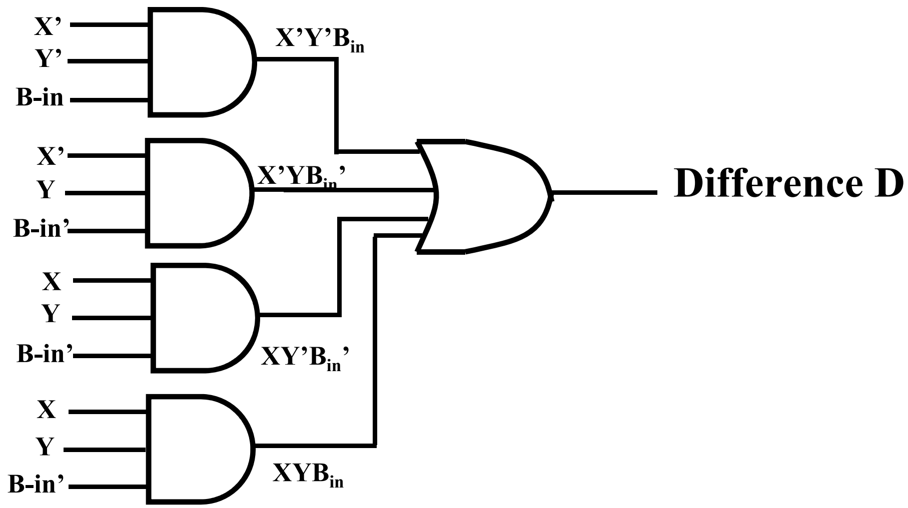
B_{in} \ XY	00	01	11	10
0	0	2 1	6	4
1	1 1	3 1	7 1	5

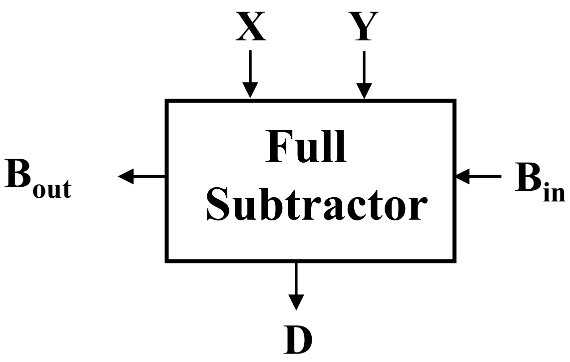
X over columns 11, 10; Y over columns 01, 11; B_{in} beside row 1.

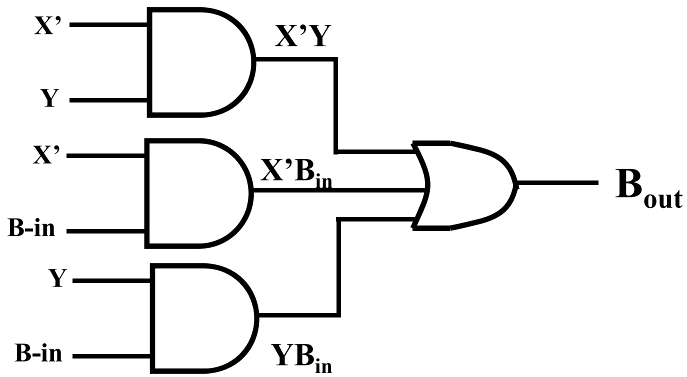
$$B_{out} = X'Y + X'(B_{in}) + Y(B_{in})$$

Full Subtractor Circuit Using AND-OR

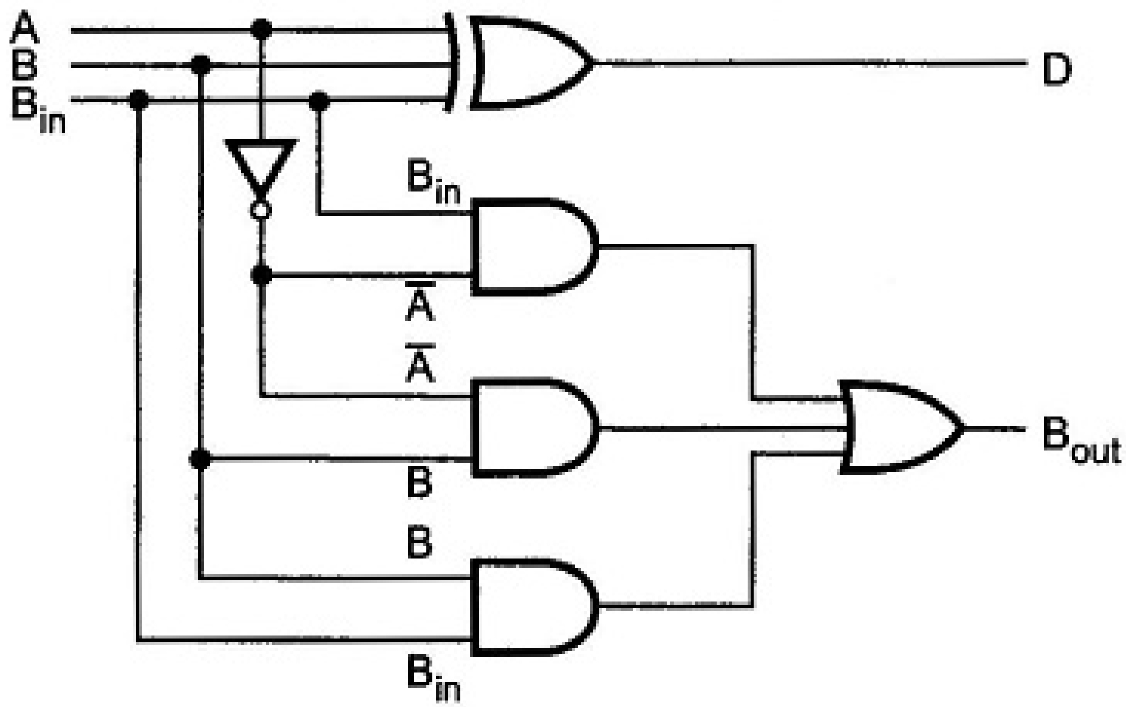








Circuit Using XOR



Implementation of N-bit Subtractors

- An n-bit subtractor used to subtract an n-bit number Y from another n-bit number X (i.e., X-Y) can be built in one of two ways:
  - By using n full subtractors and connecting them in series, creating a borrow ripple subtractor:
    - Each borrow out B-out from a full subtractor at position j is connected to the borrow in B-in of the full subtractor at the higher position j+1.
  - By using an n-bit adder and n inverters:
    - Find 2's complement of Y by:
      - Inverting all the bits of Y using the n inverters.
      - Adding 1 by setting the carry in of the least significant position to 1
    - The original subtraction (X - Y) now becomes an addition of X to two's complement of Y using the n-bit adder.

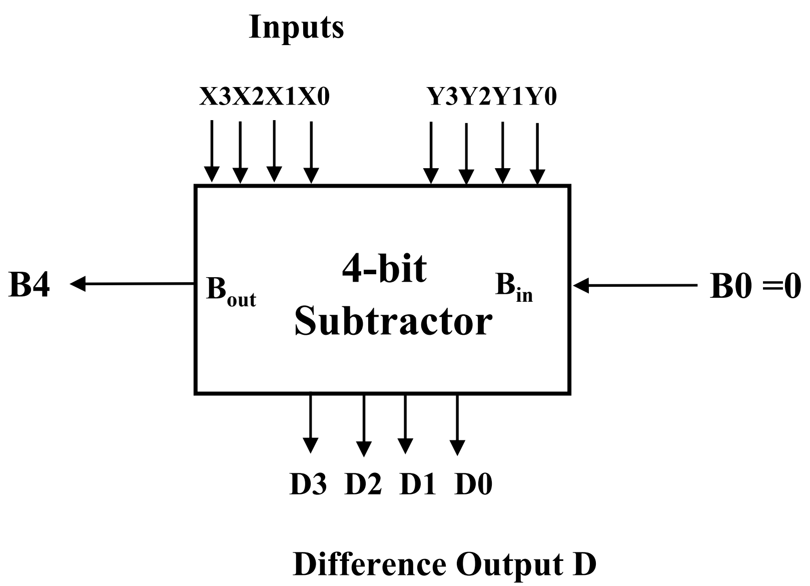
4-bit Borrow Ripple Subtractor

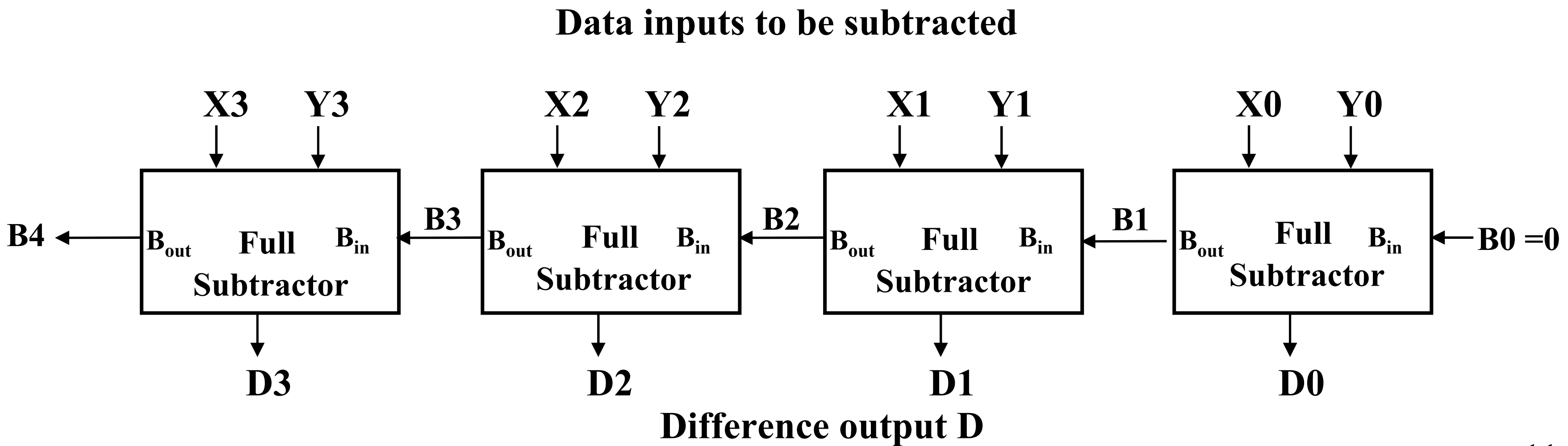
- Subtracts two 4-bit numbers:

  Y = Y3 Y2 Y1 Y0 from

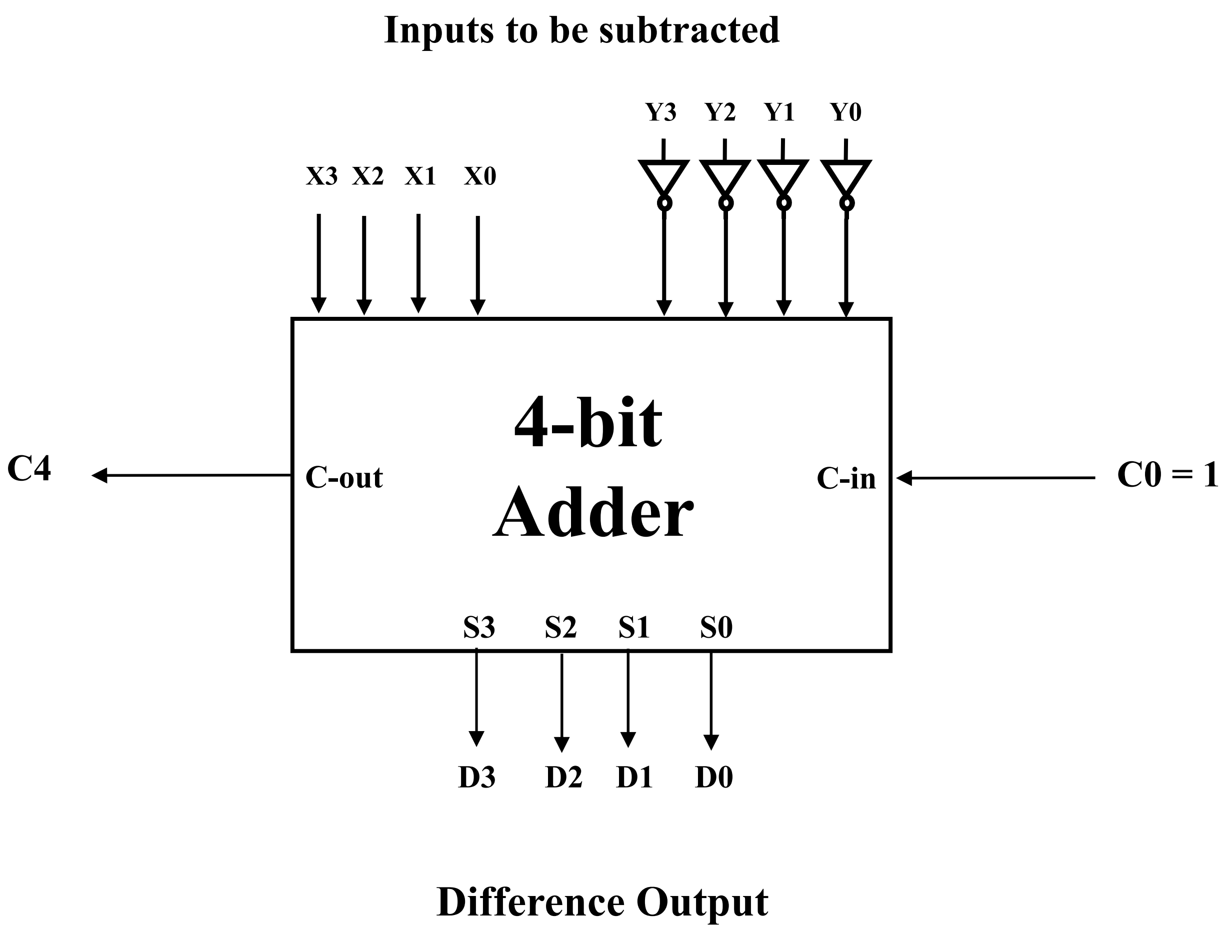
  X = X3 X2 X1 X0

  producing the difference D = D3 D2 D1 D0 , B_{out} = B4 from the most significant position j=3

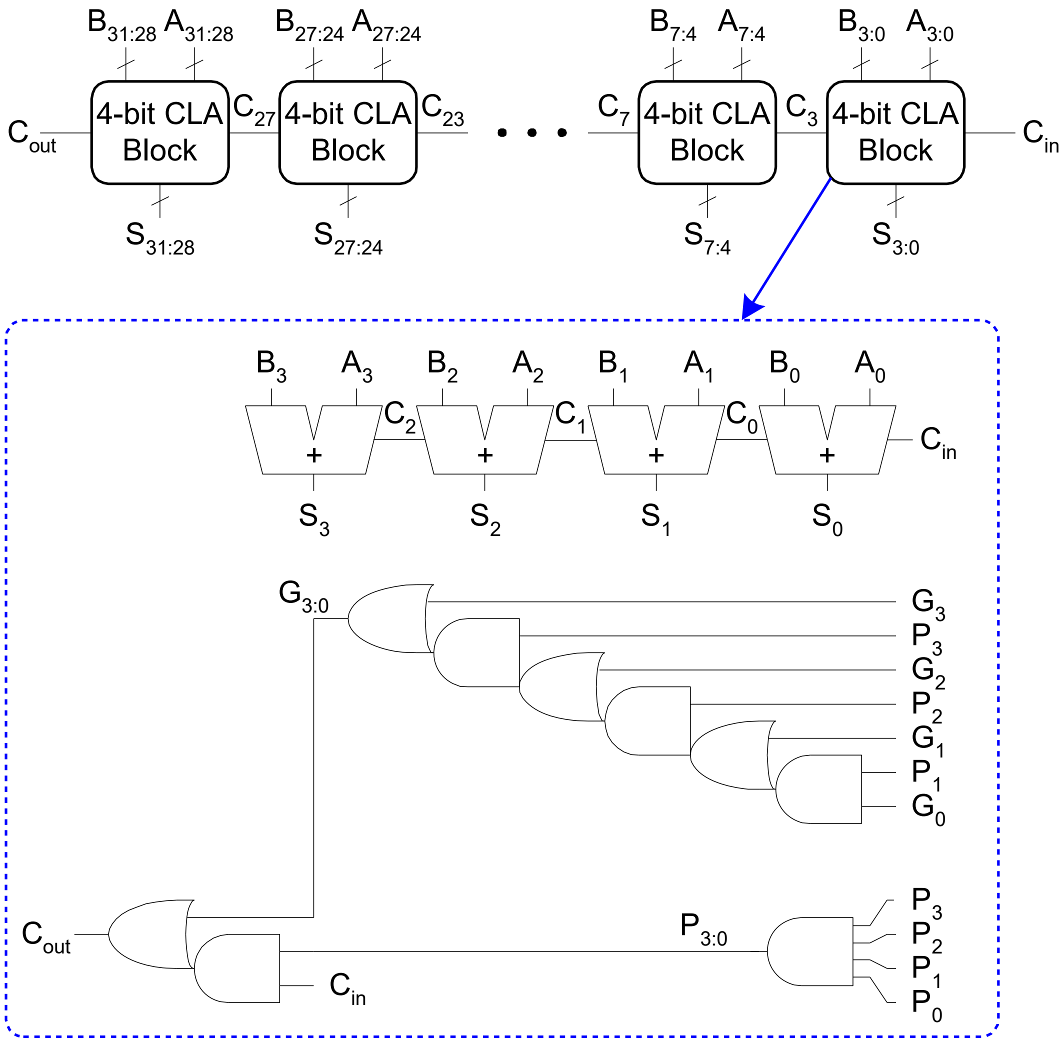




4-bit Subtractor Using 4-bit Adder



32-bit CLA with 4-bit Blocks



Carry-Lookahead Adder Delay

$$\boldsymbol{t_{CLA} = t_{pg} + t_{pg\_block} + (N/k - 1)\, t_{AND\_OR} + k t_{FA}}$$

t_{pg}: delay to generate all $P_i G_i$

t_{pg_block}: delay of generate all $P_{i:j}$, $G_{i:j}$

t_{AND_OR}: delay from C_{in} to C_{out} of final AND/OR gate in k-bit CLA block

Delay: RCA vs. CLA

- Gate levels of C_i and S_i
  - RCA (2 and 1) vs. CLA (3 and 4)
- Different metric types
  - Only gate levels
    - All gates share the same cost
  - With specs
    - Delay of typical gate X # of gate levels
    - E.g.,
      - Carry propagation delay: 12 ns
      - Sum propagation delay: 15 ns
- An N-bit carry-lookahead adder is generally much faster than a ripple-carry adder for N > 16

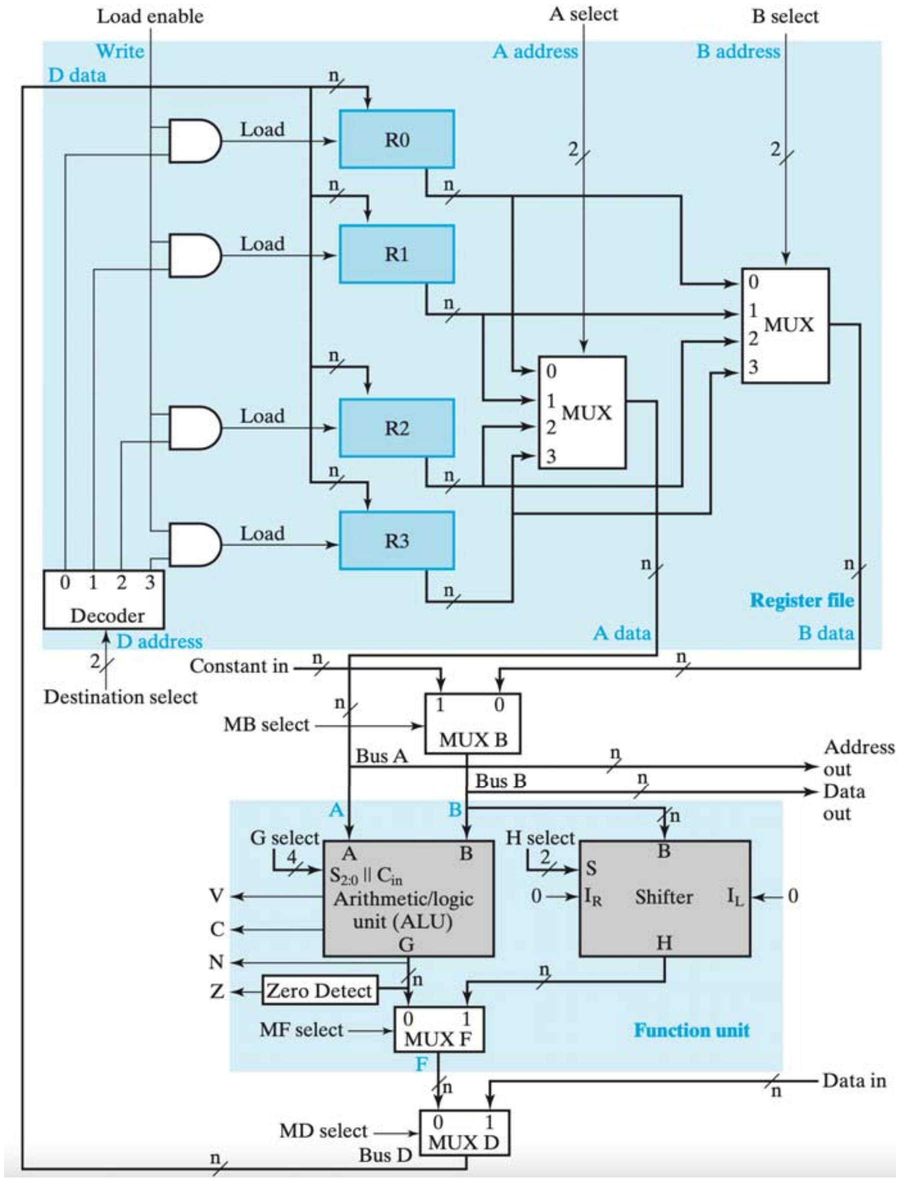
Datapath and ALU

- The datapath and the control unit are the two parts of the processor, or CPU, of a computer.

- Instead of having each individual register perform its microoperations directly, computer systems often employ many storage registers in conjunction with a shared operation unit called an arithmetic/logic unit, abbreviated **ALU**.
  - To perform a microoperation, the contents of specified source registers are applied to the inputs of the shared ALU. The ALU performs an operation, and the result of this operation is transferred to a destination register.

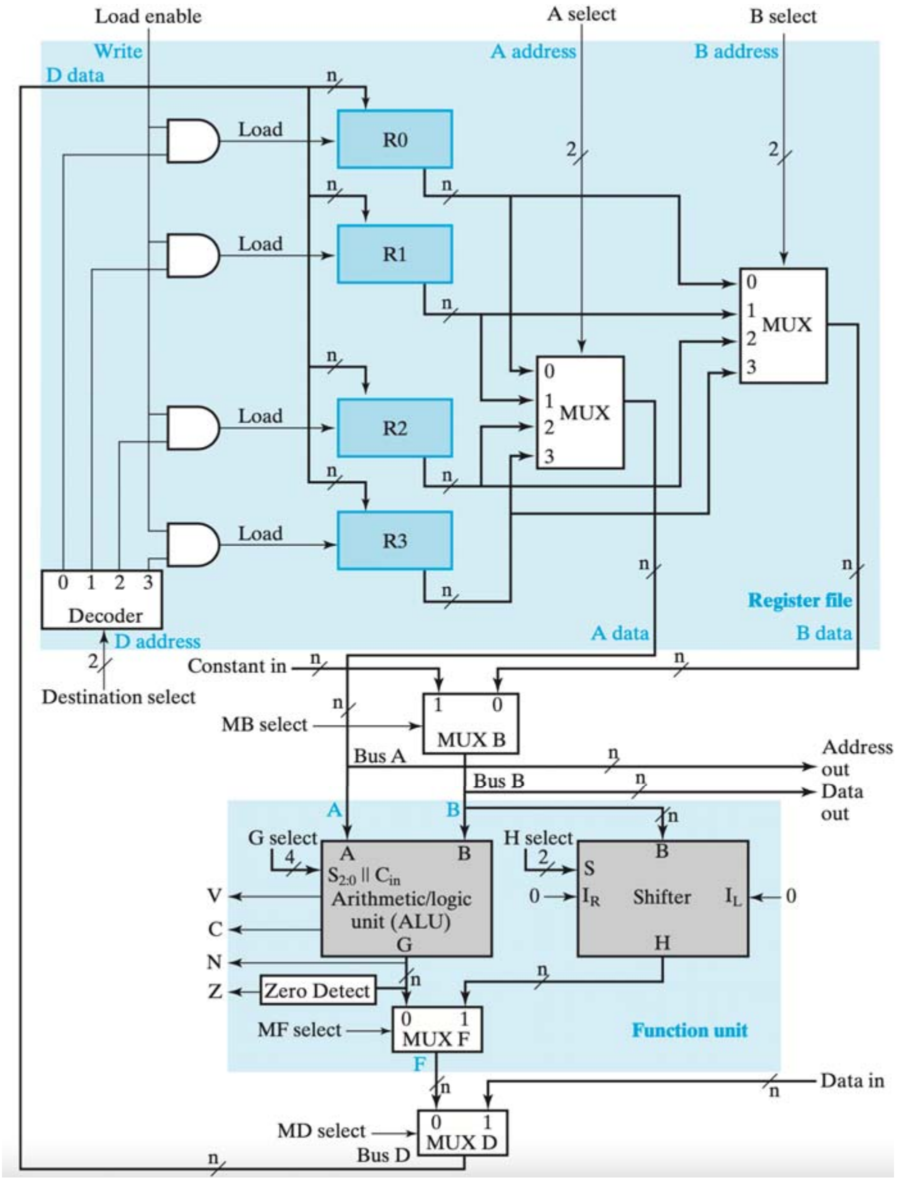
Datapath = ALU + Registers

- The combination of a set of registers with a shared ALU and interconnecting paths is the datapath for the system.
- The control unit for the datapath directs the information low through the buses, the ALU, the shifter, and the registers by applying signals to the select inputs.



ALU in the Datapath

- With the ALU as a combinational circuit, the entire register transfer operation from the source registers, through the ALU, and into the destination register is performed during one clock cycle.
    - The shift operations are often performed in a separate unit, but sometimes these operations are also implemented within the ALU.



How to Revise it?

```
     01010010 (multiplicand)
   x 01101101 (multiplier)
0000000000000000 (partial sum)
```

How to Revise it?

```
         01010010 (multiplicand)
       x 01101101 (multiplier)
       ------------------------
000000001010010 (partial sum)
```

How to Revise it?

```
       01010010 (multiplicand)
     x 01101101 (multiplier)
0000000001010010 (partial sum)
       00000000
```

How to Revise it?

```
   01010010 (multiplicand)
 x 01101101 (multiplier)
 ______________________
000000001010010 (partial sum)
```

How to Revise it?

```
      01010010 (multiplicand)
     x 01101101 (multiplier)
     ------------------------
000000001010010 (partial sum)
       01010010
```

How to Revise it?

```
   01010010 (multiplicand)
  x 01101101 (multiplier)
-----------------
000000110011010 (partial sum)
```

How to Revise it?

```
    01010010 (multiplicand)
    x 01101101 (multiplier)
    ------------------------
000000110011010 (partial sum)
    01010010
```

How to Revise it?

```
      01010010 (multiplicand)
     x 01101101 (multiplier)
0000100000101010 (partial sum)
```

How to Revise it?

```
    01010010 (multiplicand)
      x 01101101 (multiplier)
000010000101010 (partial sum)
    00000000
```

How to Revise it?

```
   01010010 (multiplicand)
     x 01101101 (multiplier)
-----------------------------
0000100000101010 (partial sum)
```

How to Revise it?

```
   01010010 (multiplicand)
       x 01101101 (multiplier)
      ________________________
 00010000101010 (partial sum)
   01010010
```

How to Revise it?

```
       01010010 (multiplicand)
          x 01101101 (multiplier)
     ___________________________
     000111001101010 (partial sum)
```

How to Revise it?

```
 01010010 (multiplicand)
      x 01101101 (multiplier)
000111001101010 (partial sum)
 01010010
```

How to Revise it?

```
  01010010 (multiplicand)
     x 01101101 (multiplier)
     ----------------------
010001011101010 (partial sum)
```

How to Revise it?

```
01010010 (multiplicand)
     x 01101101 (multiplier)
     ______________________
010001011101010 (partial sum)
00000000
```

How to Revise it?

```
01010010 (multiplicand)
      x 01101101 (multiplier)
 010001011101010 (full sum)
```

Example of Signed Multiplication

- Note:
  - Sign extension of partial product
  - If most significant bit of multiplier is 1, then subtract

$00111_2 * 0111_2$	$11001_2 * 1001_2$	$00111_2 * 1001_2$	$11001_2 * 0111_2$
00000 0111	00000 1001	00000 1001	00000 0111
+ 00111 0111	+ 11001 1001	+ 00111 1001	+ 11001 0111
→ 00011 1011	→ 11100 1100	→ 00011 1100	→ 11100 1011
+ 01010 1011	→ 11110 0110	→ 00001 1110	+ 10101 1011
→ 00101 0101	→ 11111 0011	→ 00000 1111	→ 11010 1101
+ 01100 0101	subtract	subtract	+ 10011 1101
→ 00110 0010	00110 0011	11001 1111	→ 11001 1110
→ 00011 0001	→ 00011 0001	→ 11100 1111	→ 11100 1111

Booth's Encoding

- Really just a new way to encode numbers
  - Normally positionally weighted as 2^n
  - With Booth, each position has a sign bit
  - Can be extended to multiple bits

0	**1**	**1**	**0**	**Binary**
+1	0	-1	0	1-bit Booth
+2		-2		2-bit Booth

Y_j	Y_{j-1}	**1-bit Encoding**
0	0	+0
0	1	+1
1	0	-1
1	1	-0

Y_{j+1}	Y_j	Y_{j-1}	**2-bit Encoding**
0	0	0	+0
0	0	1	+1
0	1	0	+1
0	1	1	+2
1	0	0	-2
1	0	1	-1
1	1	0	-1
1	1	1	-0

2-bits/cycle Booth Multiplier

- For every pair of multiplier bits
  - If Booth's encoding is '-2'
    - Shift multiplicand left by 1, then subtract
  - If Booth's encoding is '-1'
    - Subtract
  - If Booth's encoding is '0'
    - Do nothing
  - If Booth's encoding is '1'
    - Add
  - If Booth's encoding is '2'
    - Shift multiplicand left by 1, then add

2 bits/cycle Booth's

1 bit Booth	
00	+0
01	+M;
10	-M;
11	+0

Current	Previous	Operation	Explanation
00	0	+0;shift 2	[00] => +0, [00] => +0; 2x(+0)+(+0)=+0
00	1	+M; shift 2	[00] => +0, [01] => +M; 2x(+0)+(+M)=+M
01	0	+M; shift 2	[01] => +M, [10] => -M; 2x(+M)+(-M)=+M
01	1	+2M; shift 2	[01] => +M, [11] => +0; 2x(+M)+(+0)=+2M
10	0	-2M; shift 2	[10] => -M, [00] => +0; 2x(-M)+(+0)=-2M
10	1	-M; shift 2	[10] => -M, [01] => +M; 2x(-M)+(+M)=-M
11	0	-M; shift 2	[11] => +0, [10] => -M; 2x(+0)+(-M)=-M
11	1	+0; shift 2	[11] => +0, [11] => +0; 2x(+0)+(+0)=+0

Why It Works?

- Booth's algorithm performs an addition when it encounters the first digit of a block of ones (0 1) and a subtraction when it encounters the end of the block (1 0).
- When the ones in a multiplier are grouped into long blocks, Booth's algorithm performs fewer additions and subtractions than the normal multiplication algorithm.

$b \times (a_{31}a_{30}....a_0)$

$$
\begin{aligned}
&= a_0 \times b \times 2^0 + \\
&\quad a_1 \times b \times 2^1 + \\
&\quad \ldots \\
&\quad a_{31} \times b \times 2^{31}
\end{aligned}
$$

$$
\begin{aligned}
&= (\ 0 - a_0\ ) \times b \times 2^0 + \\
&\quad (\ a_0 - a_1) \times b \times 2^1 + \\
&\quad \ldots \\
&\quad (a_{30} - a_{31}) \times b \times 2^{31} + \\
&\quad a_{31} \times b \times 2^{32}
\end{aligned}
$$

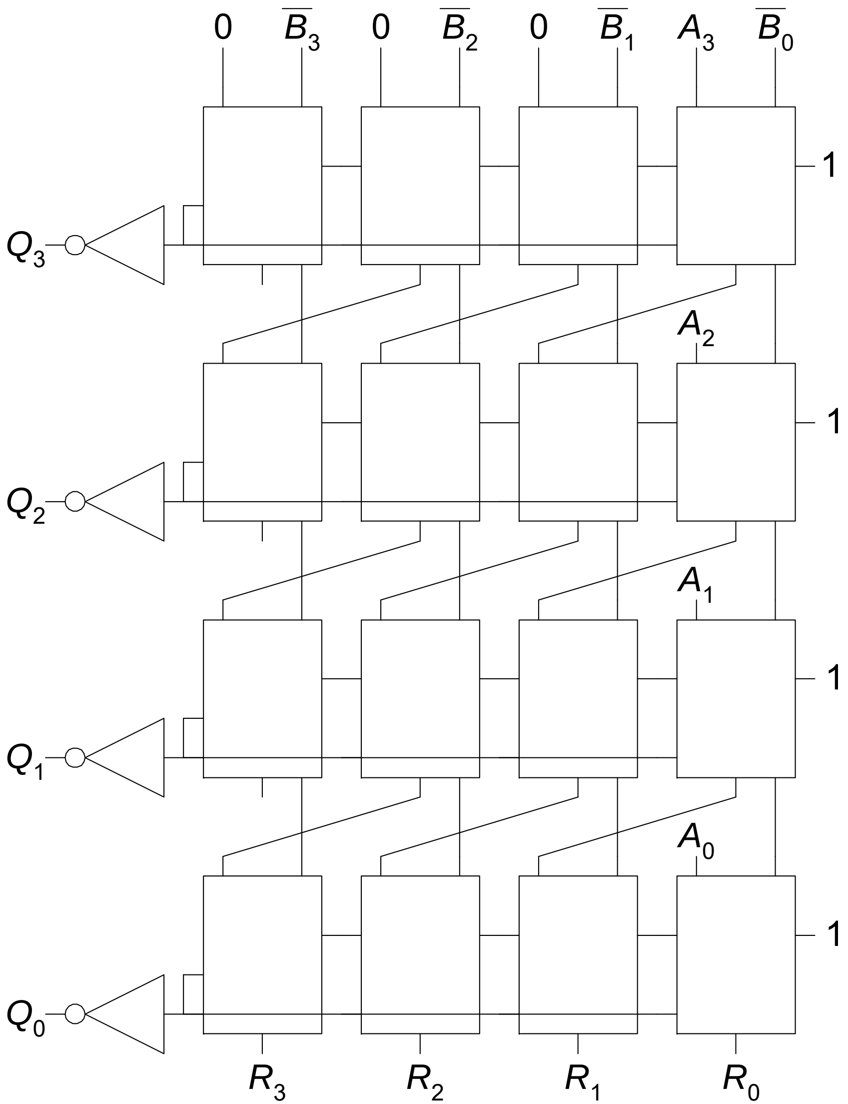
Why It Works?

- Works for positive multiplier coz a_{31} is 0
- What happens for negative multipliers? a_{31} is 1
  - Remember that we are using 2's complement binary

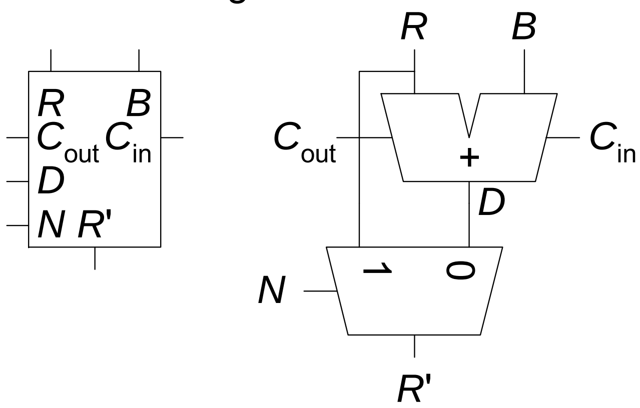
$$b \times (a_{31}a_{30}\ldots.a_0) = b \times (-\ a_{31} \times 2^{31}+a_{30} \times 2^{30}+\ldots.a_0 \times 2^0)$$

Same derivation applies

4 x 4 Divider



Legend



A/B = Q + R/B

Algorithm:

R′ = 0

for *i* = N-1 to 0

R = {R′ << 1. A_i}

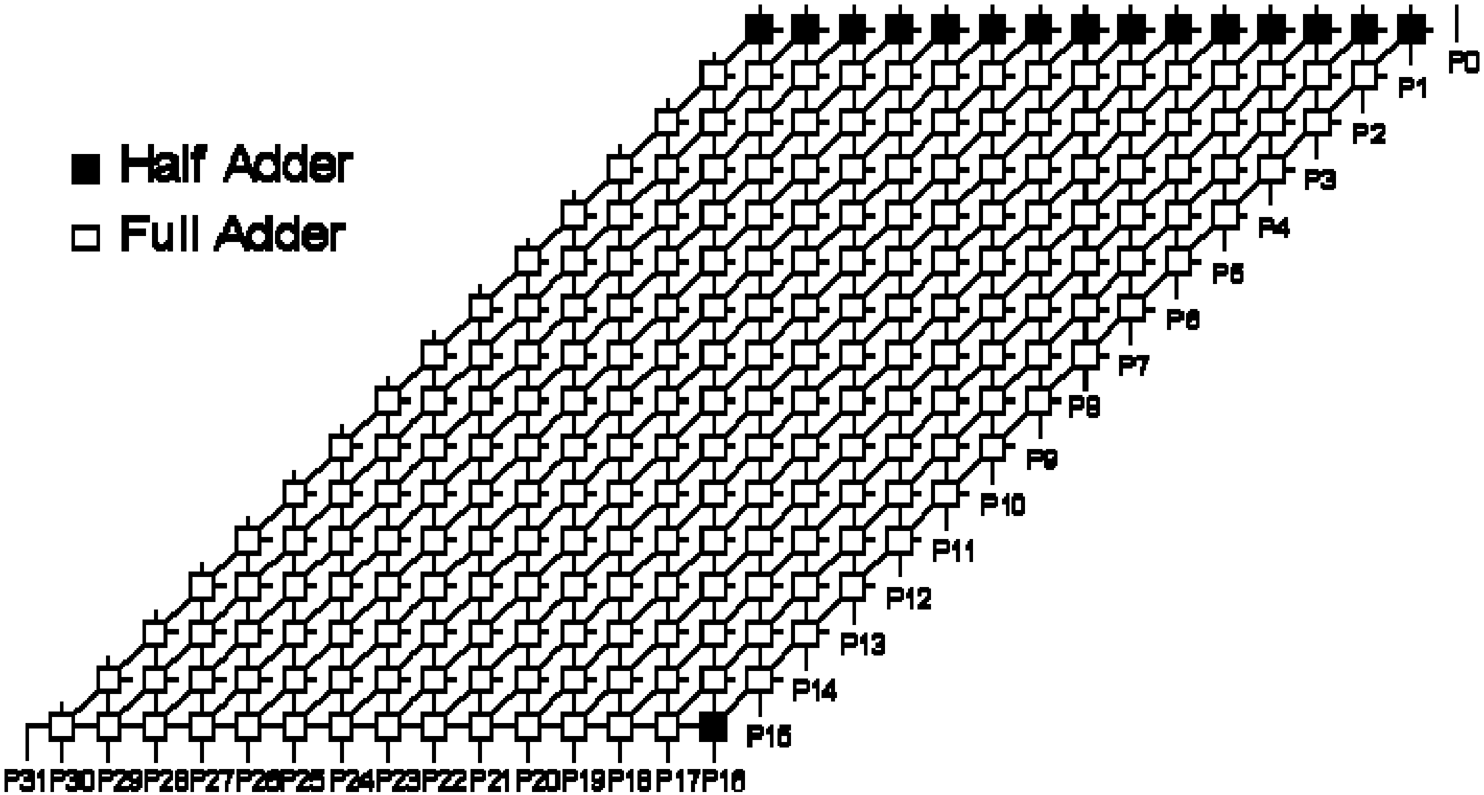
D = R - B

if D < 0, Q_i=0, R′=R

else Q_i=1, R′=D

R′=Rc

16-bit Array Multiplier



Conceptually straightforward

Fairly expensive hardware, integer multiplies relatively rare

Most used in array address calc: replace with shifts